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TA || Ashka || TM- B141

ashka singh <sgsinghashka@gmail.com>
To: Dhiraj Kr <dhiraj.aiworkspace@gmail.com>

24 September 2026 at 22:38

Hi Sir,

Thank you for your valuable feedback and encouragement. I really appreciate your guidance throughout the course.

I'm also sharing my project details in the attached Excel and PDF files, which include the **project names, technologies used, and GitHub links** for each project.

I would appreciate it if you could review my projects and let me know if there are any areas where I should make further improvements or additions on my side.

Thank you again for your support and guidance.

Regards
Ashka Singh

On Thu, 24 Sept 2026 at 21:49, Dhiraj Kr <dhiraj.aiworkspace@gmail.com> wrote:

Good Evening,

I have received your assignments.

You're performing well. Just maintain regular practice on a regular basis and continue submitting your work on time. Consistency will help you develop further step by step and become more more capable in your skills.

Keep going and keep up the excellent work!

Thanks & Best Regards,

Dhiraj Kr.

Founder | Data Scientist & Agentic AI & GenAI Developer Data Scientist

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On Thu, Sep 24, 2026 at 9:12 PM ashka singh <sgsinghashka@gmail.com> wrote:

Hi Sir,

Regarding the topics of Project – Expense Helper, Project – Grocery Assistant, and Project – Home Assistant . Kindly find the requested details below.

I'm submitting my pending assignments and projects as my final submission. Due to other ongoing commitments, I was unable to complete them within the scheduled timeline. I have already informed the Top Mentor Support Team regarding the delay.

Date: 7th Aug 2026

Topic: Project – Grocery Assistant**1. What is the purpose of the Grocery Assistant project?**

The Grocery Assistant is an application that helps users create a personalized grocery shopping plan based on their budget and selected requirements. It helps organize grocery items and manage spending efficiently.

2. How does the application create a budget-based grocery plan?

The main goal is to create a useful grocery list without exceeding the available budget.

Flow:

User Inputs → Select Items → Estimate Cost → Compare with Budget → Grocery Plan

3. What is the use of `st.multiselect()` in this project?

`st.multiselect()` is used when the user needs to select multiple values.

For example:

```
items = st.multiselect(
    "Select grocery items",
    ["Rice", "Milk", "Vegetables", "Fruits", "Bread"]
)
```

The user can select multiple grocery items at the same time.

4. Why is Session State used in the Grocery Assistant?

Session State helps maintain user information during the current session..

For example, it can store:

- Selected grocery items
- Budget
- Generated grocery plan
- Previous results

Without Session State, some values may reset whenever Streamlit reruns the script.

5. How does the download feature work in this application?

The application can convert the generated grocery plan into a downloadable format, such as text or CSV. Streamlit's `st.download_button()` is then used to allow the user to download the generated data.

Example:

```
st.download_button(  
  
    "Download Grocery Plan",  
  
    data=grocery_plan,  
  
    file_name="grocery_plan.txt"  
  
)
```

Date: 8th Aug 2026

Topic: Project – Expense Helper

1. What is the purpose of the Expense Helper project?

The Expense Helper is a Streamlit-based application that helps users record, calculate, and analyze their expenses. It can also use GenAI to provide expense-related assistance, such as summarizing or categorizing spending.

2. How is the total expense calculated in this project?

The total expense is calculated by adding the amounts of all recorded expenses.

For example:

Food ₹500

Travel ₹300

Shopping ₹700

Total ₹1500

In Python, this can be done using:

```
total_expense = sum(expenses)
```

3. What is the role of Streamlit in the Expense Helper project?

Streamlit converts the Python application into an interactive web app.

4. Why is the OpenAI API used in this application?

The OpenAI API is used to add Generative AI capabilities to the Expense Helper. It can understand user queries and generate useful responses, such as expense summaries, spending insights, or recommendations.

Flow:

User Input → Streamlit → OpenAI API → AI Model → Response

5. How can you protect the API key in a GenAI project?

The API key should never be hard-coded or uploaded to GitHub. It should be stored securely using environment variables, Streamlit secrets, or a secret manager.

For example, with Streamlit:

```
api_key = st.secrets["OPENAI_API_KEY"]
```

Also, if a key is accidentally exposed, it should be revoked and regenerated.

Date: 8th Aug 2026

Topic: Project – Home Assistant

1. What is the purpose of the Home Assistant project?

The Home Assistant project is a GenAI-based application that allows users to ask questions in natural language and get AI-generated responses.

2. Which technologies are used in the Home Assistant project?

Python handles the logic, Streamlit handles the UI, and the GenAI API generates responses.

3. How does Streamlit send a user question to the GenAI model?

When the user enters a question in the Streamlit interface, the application captures the input and sends it to the GenAI API through the Python code. The model processes the prompt and returns the generated response, which Streamlit displays to the user.

4. Why should an API key not be written directly inside the program?

If it is hard-coded in the program and the code is uploaded to GitHub or shared with others, someone could misuse the key and potentially create unauthorized API usage or charges.

5. How do you run and test the Home Assistant application?

If the application is built with Streamlit, I would typically run it using:

```
streamlit run app.py
```

Then I open the local URL shown in the terminal, enter different questions, and verify that the GenAI model returns appropriate responses.

I would test:

- Normal questions
- Empty input
- Different types of questions
- API errors
- Invalid/missing API key
- Response quality

Regards Ashka Singh

On Thu, 24 Sept 2026 at 20:52, ashka singh <sgsinghashka@gmail.com> wrote:

Hi Sir,

Regarding the topics of, Model Comparison, Time Series Forecasting, ARIMA, AI vs Generative AI, What is GenAI?, Prompt Engineering, Text Generation, Tokens in Generative AI, Bias, Privacy and AI Safety, Basic GenAI Project Without API Key, GPT-2 Basic Text Generation, RAG Basics, Level-1 GenAI Project – AI Email Improver, Level-1 GenAI Project – AI Hotel Booking Assistant, GGUF, Hugging Face, Knowledge Distillation, LoRA, Model Quantisation, MoE, Ollama, Vector Database. Kindly find the requested details below.

I'm submitting my pending assignments and projects as my final submission. Due to other ongoing commitments, I was unable to complete them within the scheduled timeline. I have already informed the Top Mentor Support Team regarding the delay.

Date: 27th June 2026

Topic: Model Comparison

1. What is Model Comparison?

Model Comparison means evaluating multiple machine learning algorithms on the same problem and dataset to see which model performs better.

Example:

Compare Linear Regression, Decision Tree, and Random Forest for predicting house prices.

Model Comparison = Compare models to find the most suitable one

2. Why do we compare Machine Learning models?

Because there is no single model that works best for every dataset.

We compare models to:

- Find better prediction performance
- Reduce errors
- Avoid overfitting
- Choose a model suitable for the business problem
- Balance performance and complexity

3. Difference between Linear Regression and Decision Tree Regression?

Linear Regression	Decision Tree Regression
Assumes a linear relationship	Can capture non-linear relationships
Uses a mathematical linear equation	Uses decision rules/splits
Simple and easy to interpret	Can model complex patterns
Sensitive to outliers	Generally less sensitive to outliers
Works well when relationship is roughly linear	Useful when relationships are non-linear

Example:

If house price increases roughly with area → Linear Regression may work well.

If price depends on complex rules involving area, location, rooms, etc. → Decision Tree may perform better.

4. What is RMSE?

RMSE (Root Mean Squared Error) measures the average size of prediction errors.

Formula:

$$\text{RMSE} = \sqrt{\text{Average of squared errors}}$$

Lower RMSE = Better model performance

Example:

- Model A → RMSE = 5
- Model B → RMSE = 10

Model A is better, assuming both are evaluated on the same target and dataset.

5. How do we choose the best model?

We would compare models using test/validation data and relevant evaluation metrics.

For regression:

- RMSE
- MAE
- R^2

For classification:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Then we would also consider overfitting, model complexity, interpretability, and business requirements.

Date: 27th June 2026

Topic: Time Series Forecasting

1. What is Time Series Forecasting?

Time Series Forecasting means using historical time-based data to predict future values.

Example: Predicting next month's sales using previous months' sales.

2. What is Time Series Data?

Time Series Data is data collected or recorded over a specific period of time, usually at regular intervals.

Examples:

- Daily sales
- Monthly revenue
- Hourly electricity consumption
- Yearly temperature

Example:

Month	Sales
Jan	10,000

Feb 12,000

Mar 15,000

3. Why is time important in Time Series Forecasting?

Time is important because the order of observations matters.

Previous values can influence future values, and the data may contain patterns such as:

- Trend → Long-term increase/decrease
- Seasonality → Repeating pattern
- Cycles → Longer-term fluctuations

4. Give two real-life examples of Time Series Forecasting.

Example 1 — Sales Forecasting:

Use previous monthly sales to predict future sales.

Example 2 — Electricity Demand:

Use historical electricity consumption to predict future demand.

Other examples include stock prices, weather, website traffic, and product demand.

5. What is the simple formula for forecasting the next value?

For a very simple forecasting approach, we can use the average of previous values:

Next Value = Average of Previous Values

Example:

Previous sales = 100, 120, 140

Forecast = $(100 + 120 + 140) / 3 = 120$

Date: 28th June 2026

Topic: ARIMA

1. What is ARIMA?

ARIMA stands for AutoRegressive Integrated Moving Average. It is a statistical model used for time series forecasting, mainly to predict future values based on past observations and past errors.

ARIMA = Time Series Forecasting Model

2. What is Time Series Forecasting?

Time Series Forecasting means using historical time-based data to predict future values.

Example:

Using previous monthly sales to predict next month's sales.

- Past data → Identify pattern → Predict future
-

3. What are **p**, **d**, and **q** in ARIMA?

ARIMA is represented as ARIMA(p, d, q):

- **p** = AutoRegressive (AR) → Number of previous values used
- **d** = Integrated (I) → Number of times data is differenced
- **q** = Moving Average (MA) → Number of previous forecast errors used

Example:

ARIMA(2,1,1) means:

- **p** = 2
- **d** = 1
- **q** = 1

* **p** = Past values | **d** = Differencing | **q** = Past errors

4. Why is differencing used in ARIMA?

Differencing is used to make a time series more stationary by removing trends or changes in the level of the series.

formula:

Differenced Value = Current Value – Previous Value

Example:

Sales: 100 → 120 → 150

Difference: 20 → 30

5. Where is ARIMA used?

ARIMA is commonly used for short- to medium-term forecasting when historical time-based patterns are useful.

Examples:

- Sales forecasting
- Demand forecasting

- Revenue forecasting
- Electricity consumption
- Website traffic
- Inventory planning

Date: 28th July 2026

Topic: AI vs Generative AI

1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the technology that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, understanding language, and recognizing patterns.

Examples: Chatbots, recommendation systems, fraud detection, face recognition, self-driving systems.

2. What is Generative AI?

Generative AI is a type of AI that can create new content based on the patterns it has learned from existing data.

It can generate text, images, code, audio, and video.

Examples: ChatGPT generating text, image generators creating pictures, AI coding assistants generating code.

3. What is the main difference between AI and Generative AI?

AI	Generative AI
Broad field of intelligent systems	Subset of AI
Can predict, classify, detect, recommend, etc.	Focuses on generating new content
May not create anything new	Creates text, images, code, audio, etc.
Example: Spam detection	Example: ChatGPT generating an email

4. Can an AI system work without generating new content?

Yes. AI does not necessarily need to generate new content.

For example, a spam detection AI can simply classify an email as Spam or Not Spam. A fraud detection system can classify a transaction as Fraud or Not Fraud without creating any new content.

5. Why is every Generative AI system considered AI, but every AI system is not Generative AI?

Because Generative AI is a subset of Artificial Intelligence.

Think of it like:

Artificial Intelligence (AI)

↓

Generative AI

↓

Generates new content

So, Generative AI uses AI techniques, but many AI systems only predict, classify, detect, or make decisions and don't generate new content.

Date: 28th July 2026

Topic: What is GenAI?

1. What is Generative AI?

Generative AI (GenAI) is a type of Artificial Intelligence that can create new content based on patterns learned from existing data.

It can generate text, images, code, audio, and video.

2. What type of content can GenAI create?

GenAI can generate different types of content:

-  Text – emails, articles, summaries
-  Images – illustrations, designs, photos
-  Code – Python, SQL, Java, etc.
-  Audio – music, voice, sound
-  Video – short videos, animations

3. What is a prompt in Generative AI?

A prompt is the instruction or input we give to a Generative AI model to get the desired output.

Example:

"Explain Machine Learning in simple words with a real-life example."

The AI uses this instruction to generate the response.

4. How is GenAI different from a traditional search engine?

GenAI	Traditional Search Engine
Generates a response/content	Finds and ranks existing web pages
Can create new text, images, code, etc.	Mainly provides links/results
Responds conversationally	Primarily keyword/query based
Can summarize or transform information	Retrieves information from indexed sources

Example:

If you ask GenAI "Write an email requesting leave," it can create the email.

A search engine would generally find websites/templates related to leave emails.

5. Why should GenAI-generated information be verified before using it?

Because GenAI can sometimes produce incorrect, outdated, incomplete, or fabricated information, even when the response sounds confident. This is often referred to as an AI hallucination.

For important decisions, we should verify information using reliable and authoritative sources.

Example:

If GenAI gives a medical, legal, financial, or technical claim, we should verify it before acting on it.

Date: 28th July 2026

Topic: Prompt Engineering

1. What is Prompt Engineering?

Prompt Engineering is the process of designing and improving instructions given to a Generative AI model to get accurate, relevant, and consistent responses.

Example:

Instead of: *"Tell me about sales."*

Use: *"Act as a data analyst. Analyze the sales data and identify the top 3 trends in a table."*

2. What are the main elements of a good prompt?

The main elements are:

1. Role – Who should the AI act as?
2. Task – What should it do?
3. Context – What background information is needed?
4. Constraints – What limitations should it follow?
5. Examples – Show expected input/output when useful.
6. Output Format – Specify how the answer should look.

3. What is the difference between Zero-Shot and Few-Shot Prompting?

Zero-Shot	Few-Shot
No examples are provided	Multiple examples are provided
AI understands the task from instructions	AI learns the expected pattern from examples
Simple tasks	Useful for specific formats or complex patterns

Example:

Zero-shot:

"Classify this review as Positive or Negative."

Few-shot:

"Good product → Positive
 Very bad service → Negative
 Amazing quality → Positive
 Now classify: 'The product was excellent.'"

I

4. Why can a detailed prompt still generate an incorrect response?

A detailed prompt does not guarantee a correct answer.

The response can still be incorrect because of:

- Incorrect or incomplete context
- Ambiguous instructions
- Incorrect assumptions by the model
- Poor-quality input data
- Model limitations
- Outdated information
- Hallucinations

5. How would you improve a prompt that produces a general and unrelated answer?

I would make the prompt more specific and structured.

Example

 Poor prompt:

"Tell me about customer data."

 Improved prompt:

"Act as a data analyst. Analyze customer purchase data and identify the top 3 customer segments based on purchase frequency and total spending. Present the results in a table and give 2 business recommendations."

This adds role, specific task, context, criteria, and output format.

Date: 28th July 2026

Topic: Text Generation

1. What is Text Generation in Generative AI?

Text generation is the process where a Generative AI model creates new text based on a user's prompt or instructions.

It can generate emails, articles, summaries, stories, answers, reports, and code.

2. What is a Prompt in Text Generation?

A prompt is the input or instruction given to the AI model that tells it what to generate and how to generate it.

Example:

"Write a professional email requesting two days of leave."

3. How does an AI Model Generate Text?

At a high level:

Prompt → Understand context → Predict next tokens → Generate sequence → Final text

The model predicts the most likely next token based on the prompt and previously generated tokens. It continues this process until it produces the response.

Example:

Input: "The sky is"

The model may predict "blue" as a likely next token.

4. Why does a Detailed Prompt Usually Produce a Better Response?

A detailed prompt gives the model clearer instructions, context, constraints, and expected output format.

For example:

✗ "Explain Python."

✓ "Explain Python to a beginner in simple language, give 3 real-life examples, and keep the answer under 100 words."

The second prompt reduces ambiguity, so the response is usually more relevant.

5. Can an AI Model Generate Grammatically Correct Text That Contains Incorrect Information?

Yes. An AI model can produce text that is well-written and grammatically correct but factually incorrect.

This can happen because language models are designed to generate likely and coherent text; they do not automatically guarantee that every statement is factually true.

Example:

An AI might confidently provide an incorrect historical date while writing it in perfect English.

Date: 28th July 2026

Topic: Tokens in Generative AI

1. What is a Token in Generative AI?

A token is a small unit of text that an AI model processes. A token can be a word, part of a word, a character, or punctuation, depending on the tokenizer.

Example:

I love data science! may be broken into several tokens, such as words and punctuation.

2. What is Tokenisation?

Tokenisation is the process of breaking text into smaller units called tokens so that the AI model can process them. The exact token breakdown depends on the model's tokenizer.

Flow:

Text → Tokenisation → Tokens → AI Model

For example:

I am learning Python.

↓

I | am | learning | Python | .

3. How does an AI Model utilise Input and Output Tokens?

An AI model processes the input tokens from the prompt and generates output tokens for the response.

Flow:

User Prompt → Input Tokens → Model Processing → Output Tokens → Response

For example:

Prompt: "Explain Machine Learning."

The prompt is converted into input tokens. The model then generates output tokens that form the explanation.

Token counts are also commonly relevant to context limits and API usage/costs.

4. What is the Formula for Calculating Total Tokens?

The formula is:

Total Tokens = Input Tokens + Output Tokens

Example:

- Input = 100 tokens
- Output = 300 tokens

Total = 100 + 300 = 400 tokens

For an API interaction, other token categories such as cached or reasoning tokens may sometimes be reported separately, depending on the model/provider.

5. Why Can the Same Sentence Use Different Token Counts in Different AI Models?

Because different AI models can use different tokenisation methods and vocabularies.

One model might represent a word as one token, while another might split the same word into multiple tokens.

Tokenisation can also vary based on:

- Language
- Punctuation
- Special characters
- Numbers
- Model vocabulary

Date: 29th July 2026

Topic: Bias, Privacy and AI Safety

1. What is Bias in Artificial Intelligence?

AI bias occurs when an AI system produces systematically unfair, inaccurate, or unbalanced results for certain groups or situations.

Bias can come from the training data, model design, or human decisions used to build the system.

Example:

If an employee-promotion model is trained on historically biased data, it may unfairly favor certain groups.

2. How Can Bias Enter an AI System?

Bias can enter at different stages:

- Biased training data – data doesn't represent all groups fairly.
- Historical bias – past human decisions contain unfair patterns.
- Sampling bias – some groups are underrepresented.
- Human/model design choices – features, labels, or objectives may introduce bias.

- Feedback loops – biased predictions can influence future data.
-

3. Why Should Users Avoid Sharing Confidential Data with AI Tools?

Users should avoid entering confidential information because sensitive data may be stored, processed, logged, or exposed depending on the AI tool and its privacy settings.

Examples of information to avoid sharing unnecessarily:

- Passwords and authentication credentials
- Credit/debit card details
- Personal identification information
- Confidential company documents
- Proprietary business information

For workplace use, users should follow their organization's AI and data-security policies.

4. What is AI Safety?

AI safety is the practice of designing, testing, and using AI systems so they operate reliably, securely, and within appropriate boundaries, while reducing harmful or unintended outcomes.

It includes:

- Preventing harmful outputs
 - Protecting user data
 - Testing model behavior
 - Reducing bias
 - Monitoring AI systems
 - Adding appropriate human oversight
-

5. Can an AI System Give an Unsafe Result Even When Its Answer Sounds Correct?

Yes. An AI response can sound confident, logical, and grammatically correct but still be incorrect or unsafe.

This can happen because AI models can make mistakes, misunderstand context, or generate unsupported information.

Example:

An AI gives incorrect medical or financial advice that sounds convincing. Acting on it without verification could cause harm.

Date: 29th July 2026

Topic: Basic GenAI Project Without API Key

1. What is the purpose of this GenAI project?

The purpose is to build a Generative AI application that can generate text using a pre-trained AI model without depending on a paid external API.

For example, a user enters a prompt like:

"Explain Machine Learning in simple words."

The local GenAI model processes the prompt and generates a response.

2. How can we use a GenAI model without an API key?

We can download and run a pre-trained open-source model locally using libraries such as Hugging Face Transformers.

Basic flow:

Transformers → Download Model → Load Locally → Give Prompt → Generate Response

Since the model runs on our own machine, an external AI API key is not required.

3. What is the use of the Transformers **pipeline**?

The **pipeline** function from Hugging Face Transformers provides a simple, high-level interface for performing common AI tasks.

For text generation:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
result = generator("Artificial Intelligence is")
```

It handles much of the underlying work, such as loading the model/tokenizer and preparing inputs and outputs.

4. Why is the internet required during the first execution?

During the first execution, the required model files, tokenizer files, and configuration files may need to be downloaded from the model repository, such as Hugging Face Hub.

First run:

Internet → Download model → Save/cache locally

Later runs:

Local cache → Load model → Generate output

So, internet is generally required for the initial download, but not necessarily for subsequent offline execution if all required files are already cached locally.

5. What is the difference between running a Local Model and using an AI API?

Local Model	AI API
Model runs on your machine	Model runs on provider's servers
Usually no API key required	Usually requires API key/authentication
Internet may not be needed after download	Internet required for requests
Uses local CPU/GPU/RAM	Provider provides computing infrastructure
More control over model/data	Easier to access powerful hosted models
Can be slower on limited hardware	Often faster depending on service/network

Simple Example

Local:

Your App → Local Model → Response

API:

Your App → Internet → AI Provider → Response

Date: 29th July 2026

Topic: GPT-2 Basic Text Generation

1. What is GPT-2?

GPT-2 (Generative Pre-trained Transformer 2) is a pre-trained language model developed by OpenAI that can generate human-like text based on a given input.

It is mainly designed for text generation, not as a modern instruction-following chatbot.

2. How does GPT-2 generate text?

GPT-2 generates text one token at a time.

Flow:

Prompt → Tokenization → Predict next token → Add token → Repeat → Generated text

For example:

Prompt: "Artificial Intelligence is"

GPT-2 predicts a likely next token such as "a", then continues predicting subsequent tokens based on the context.

3. What is the use of the **text-generation** pipeline?

The Hugging Face **text-generation** pipeline provides a simple way to use a language model for generating text without manually handling all the tokenizer and model-processing steps.

Example:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

```
result = generator("Machine learning is")
```

```
print(result)
```

4. What does **max_new_tokens** control?

max_new_tokens controls the maximum number of new tokens the model can generate after the input prompt.

Example:

```
generator(  
    "Artificial Intelligence is",  
    max_new_tokens=50  
)
```

This allows the model to generate up to 50 new tokens.

It does not count the tokens already present in the prompt.

5. Why is GPT-2 not suitable as a reliable Question-Answering chatbot?

GPT-2 was primarily designed for text continuation/generation, not reliable factual question answering or modern instruction following.

It can:

- Generate fluent responses
- Continue text based on context
- Produce plausible-sounding answers

But it can also:

- Generate incorrect or fabricated information
- Struggle with complex instructions
- Lack reliable grounding in external/current information
- Produce inconsistent answers

Date: 29th July 2026

Topic: RAG Basics

1. What is RAG in Generative AI?

RAG (Retrieval-Augmented Generation) is a technique that combines information retrieval with an LLM to generate answers using relevant external documents or data.

Instead of relying only on what the model learned during training, RAG first retrieves relevant information and then uses it to generate the answer.

2. Why do we use RAG with an LLM?

We use RAG to make LLM responses more relevant, grounded, and useful with specific or private information.

Common use cases:

- Company documents
- PDFs and reports
- Product documentation
- Customer-support knowledge bases
- Frequently updated information

Example:

A company can build a chatbot that answers questions from its internal HR policies without retraining the LLM on

those documents.

3. How does a RAG system generate an answer?

The basic process is:

User Question

↓

Convert question into a query/embedding

↓

Retriever searches relevant documents

↓

Relevant chunks are retrieved

↓

LLM receives question + retrieved context

↓

LLM generates answer

Example:

Question: *"What is our company's leave policy?"*

→ RAG retrieves the relevant HR policy section

→ LLM uses that section to generate the answer.

4. What is the role of a Retriever in RAG?

The retriever finds the most relevant information from the available knowledge source based on the user's question.

It may search using:

- Keyword-based search
- Vector/semantic search
- Hybrid search

For example, if the user asks about the refund policy, the retriever should find the document chunks related to refunds rather than unrelated documents.

5. Can a RAG system provide an incorrect answer even when the correct information exists in its documents?

Yes. The presence of correct information in the documents does not guarantee a correct answer.

Possible reasons include:

- The retriever fails to find the correct document/chunk.
- Retrieved chunks don't contain enough context.
- Poor document chunking.

- Incorrect or outdated source documents.
- The LLM misunderstands or misuses the retrieved context.
- Too much irrelevant context distracts the model.

Example:

Documents contain: Correct refund policy 

Retriever returns: An unrelated cancellation policy 

LLM: Generates an answer based on the wrong context 

Date: 4th Aug 2026

Topic: Level-1 GenAI Project – AI Email Improver

1. What is an AI Email Improver?

An AI Email Improver is an application that uses Generative AI to rewrite and improve an email by making it more professional, clear, grammatically correct, and well-structured.

Example:

Input → Informal email

Output → Professional and polished email

2. What is the role of the Gemini API in this project?

The Gemini API connects the application to Google's Gemini Generative AI model.

It takes the user's email and prompt as input, sends them to Gemini, and receives the improved email as the generated response.

Flow:

Email → Prompt → Gemini API → Gemini Model → Improved Email

3. Why is `getpass()` used for the API key?

`getpass()` is used to securely enter the API key without displaying it on the screen while typing.

Example:

```
from getpass import getpass
```

```
api_key = getpass("Enter API Key: ")
```

This is safer than directly writing the key in the source code.

4. What is the purpose of the Prompt?

The prompt tells the Gemini model what task to perform and how the output should be generated.

For example:

"Rewrite this email professionally, correct grammar, and maintain the original meaning."

A good prompt helps produce a more relevant, consistent, and controlled response.

5. What does **response.text** return?

response.text returns the text content generated by the Gemini model from the API response.

For example:

```
response = model.generate_content(prompt)
```

```
print(response.text)
```

The output would be the improved email text generated by Gemini.

Date: 4th Aug 2026

Topic: Level-1 GenAI Project – AI Hotel Booking Assistant

1. What is an AI Hotel Booking Assistant?

An AI Hotel Booking Assistant is a Generative AI application that interacts with guests in a conversational way and helps them collect booking details, answer questions, and guide them through the booking process.

2. What information does this project collect from a guest?

Typically, the assistant collects information such as:

- Guest name
- Check-in date
- Check-out date
- Number of guests
- Room type/preference
- Number of rooms
- Special requests, if applicable

Main details = Guest + Dates + Guests + Room requirements.

3. What is the role of the Prompt in this project?

The prompt provides instructions to the AI about how to interact with the guest and what information to collect.

For example, the prompt can tell the AI to:

- Ask for missing booking details.
- Respond politely.
- Summarize the guest's requirements.
- Avoid making unsupported claims.

Prompt = Controls the assistant's behavior and conversation flow.

4. Why should the AI not say that the room is confirmed?

AI can collect booking details, but confirmation requires real-time availability and booking-system integration.

5. What features are required to convert this project into a real booking system?

A real booking system would need:

1. Hotel inventory/availability API – check real-time room availability.
2. Booking database – store reservations securely.
3. Reservation API/backend – create, modify, and cancel bookings.
4. Payment gateway – process payments securely.
5. Authentication – protect guest/admin accounts.
6. Email/SMS confirmation – send actual booking confirmations.
7. Cancellation/refund handling – manage booking changes.
8. Human support/escalation – handle complex cases.

Real system flow:

Guest → AI Assistant → Availability API → Booking System → Payment → Confirmation

Date: 7th Aug 2026

Topic: GGUF

1. What is GGUF?

GGUF = A model file format commonly used for running LLMs locally.

2. Why is GGUF used in Local AI?

It supports efficient storage and loading of models and works well with CPU/GPU-based local inference.

It makes it easier to download, manage, and run compatible LLMs on personal computers.

3. Which tools commonly use GGUF Models?

GGUF models are commonly used with tools based on the llama.cpp ecosystem, including:

- llama.cpp
 - LM Studio
 - GPT4All
 - KoboldCpp
 - Other local LLM applications that support llama.cpp-compatible models
-

4. What is the Relation Between GGUF and Quantisation?

GGUF and quantisation are related, but they are not the same thing.

- GGUF → Model file/container format.
- Quantisation → Technique that reduces the numerical precision of model weights to reduce memory usage and improve inference efficiency.

A GGUF file can contain a quantised model, such as 4-bit (Q4) or 8-bit (Q8) versions.

Easy example:

Original model → Quantisation → Smaller model → Stored/distributed as GGUF

5. Why Can a GGUF Model Be Easier to Run on a Laptop Than a Full-Size Model?

Because GGUF models are often distributed in quantised versions, which can significantly reduce the model's memory requirements.

For example:

Full-precision model

→ Large memory requirement ❌

Quantised GGUF model

→ Smaller size + lower memory requirement ✅

→ More practical on a laptop

This can make local inference possible on systems with limited RAM/VRAM, although performance and quality depend on the model and quantisation level.

Date: 7th Aug 2026

Topic: Hugging Face

1. What is Hugging Face?

Hugging Face is an AI/ML platform and company that provides open-source tools, models, datasets, and libraries for building and using machine learning and Generative AI applications.

It is especially popular for NLP, LLMs, computer vision, and audio models.

2. What is Hugging Face Hub?

Hugging Face Hub is an online platform where developers can find, download, share, and upload AI models, datasets, and demos.

For example, you can search for a pre-trained model such as GPT-2, download it, and use it locally through Transformers.

3. What is a Pre-trained Model?

A pre-trained model = Already trained model that can be reused.

4. What is the use of the Transformers Library?

Transformers is a Hugging Face library that provides tools to load, use, train, and fine-tune pre-trained Transformer-based models.

It supports tasks such as:

- Text generation
- Text classification
- Question answering
- Summarization
- Translation
- Sentiment analysis

Example:

```
from transformers import pipeline
```

```
generator = pipeline("text-generation", model="gpt2")
```

5. What is the difference between using a Pre-trained Model Directly and Fine-tuning it?

Pre-trained Model Directly	Fine-tuning
Use the existing model as it is	Further train the model
No task-specific training required	Uses task/domain-specific data
Faster to implement	Requires additional training
Good for general tasks	Better for specialized tasks

Example:

Suppose we have a pre-trained language model.

Direct use:

→ Ask it to generate general text.

Fine-tuning:

→ Train it further using company-specific customer-support conversations so it performs better for that particular task.

Date: 7th Aug 2026

Topic: Knowledge Distillation

1. What is Knowledge Distillation?

Knowledge Distillation is a technique where a smaller model learns from a larger, more capable model.

2. What is a Teacher Model?

A Teacher Model is the larger or more capable model that provides knowledge or guidance to another model during training.

It usually has:

- More parameters

- Higher computational requirements
- Better performance

Example:

A large LLM can act as the Teacher.

3. What is a Student Model?

A Student Model is the Smaller model that learns from the Teacher.

The goal is to make the Student model:

- Smaller
 - Faster
 - Less memory-intensive
 - Easier to deploy
-

4. Why is Knowledge Distillation used in Generative AI?

It is used to make AI models more efficient and easier to deploy while trying to maintain much of the larger model's performance.

Benefits include:

- ⚡ Faster inference
- 📁 Lower memory usage
- 💰 Lower computational cost
- 📱 Better suitability for edge devices/laptops
- 🚀 Easier deployment

Example:

A large LLM may be too expensive to run on a laptop. Knowledge distillation can help create a smaller model that performs similar tasks more efficiently.

5. Can a Smaller Student Model Give Results Close to a Larger Teacher Model?

Yes, it can, but not always.

A well-trained Student model can achieve performance close to the Teacher on specific tasks, but some accuracy or capability may be lost because the Student has fewer parameters and less capacity.

Date: 7th Aug 2026

Topic: LoRA

1. What is LoRA?

LoRA stands for **Low-Rank Adaptation**.

It is a method used to train an existing AI model without training the complete model again.

It helps save time, cost, and computer resources.

2. Why is LoRA used in Generative AI?

LoRA helps make customization more practical by:

- Reducing trainable parameters
- Reducing memory requirements
- Making training faster
- Lowering computational cost
- Making fine-tuning more accessible

Example:

We can adapt an existing LLM for customer support or a specific writing style without retraining the entire model.

3. How does LoRA save training resources?

LoRA saves resources by training a small set of adapter parameters instead of the complete model.

4. What is the difference between LoRA and Full Fine-Tuning?

LoRA	Full Fine-Tuning
Trains small adapter parameters	Updates most/all model parameters
Lower memory requirement	Higher memory requirement
Faster and cheaper	More expensive and resource-intensive

Original weights remain frozen

Original weights are modified

Easy to maintain multiple task-specific adapters

Each customized model can require a full set of weights

5. Can LoRA customize an existing AI model without training the complete model?

Yes. That's one of the main advantages of LoRA.

The original pre-trained model remains frozen, while LoRA trains small adapter parameters that learn the required task or domain.

Example:

General LLM + LoRA adapter trained on medical text
→ More specialized for a medical-text task.

The base model is not retrained from scratch.

Date: 7th Aug 2026

Topic: Model Quantisation

1. What is Model Quantisation?

Model Quantisation is the process of reducing model precision to reduce memory and computation.

The main goal is to make the model smaller, faster, and less memory-intensive while maintaining acceptable performance.

2. Why is Quantisation used in Generative AI?

Large AI models can require significant RAM/VRAM and computing power.

Quantisation helps by:

- 📦 Reducing model size
- ⚡ Improving inference efficiency
- 🧠 Reducing memory usage
- 🛠 Lowering hardware requirements
- 💻 Making local LLMs more practical

Example:

A large LLM that requires a high-end GPU in full precision may be easier to run locally after quantisation.

3. What is the difference between FP16, INT8 and INT4?

These represent different numerical precisions:

Format	Meaning	Precision/Size	Typical Use
FP16	16-bit Floating Point	Higher	Faster AI computation while retaining good precision
INT8	8-bit Integer	Lower	Efficient inference
INT4	4-bit Integer	Much lower	Very memory-efficient local LLM inference

4. Why can a Quantised Model run better on a Laptop?

Quantised model

- Smaller memory requirement 
- More practical on CPU/GPU laptops

5. Can reducing a Model too much affect its Response Quality?

Yes. Aggressive quantisation can cause some loss in model quality because the original numerical values are represented with less precision.

Possible effects:

- Less accurate answers
- Reduced reasoning capability
- More generation errors
- Changes in language quality

Date: 7th Aug 2026

Topic: MoE

1. What is MoE in Generative AI?

MoE (Mixture of Experts) is an architecture where a large AI model is divided into multiple smaller expert networks, and only some experts are activated for each input.

2. Why is MoE used in Large AI Models?

MoE increases model capacity without requiring all parameters to be active for every input.

3. What is an Expert in MoE?

An expert is a smaller neural network or specialized component inside an MoE model.

4. What is the Role of a Router in MoE?

Router = Decides which experts are activated.

Flow:

Input Token → Router → Select Relevant Experts → Process → Combine Results

For example, the router may select the top 2 experts for a particular token instead of activating all available experts.

5. Why Does MoE Not Use All Experts for Every Input?

Because the main advantage of MoE is conditional computation.

If all experts were activated for every input, the model would require much more computation and would lose much of the efficiency advantage of MoE.

Date: 7th Aug 2026

Topic: Ollama

1. What is Ollama and why is it used in Generative AI?

Ollama is a tool that makes it easy to download and run LLMs locally on your computer.

It is used in Generative AI to run models such as Llama, Gemma, Mistral, and other supported models without sending prompts to an external AI service.

Example:

```
ollama run llama3
```

2. What is the difference between **ollama pull** and **ollama run**?

ollama pull

ollama run

Downloads a model

Runs/starts a model

Mainly used to get the model locally

Used to interact with the model

Doesn't start a chat by itself

Loads the model and lets you send prompts

3. What is the difference between using Ollama locally and using an external AI API?

Ollama:

Your App → Local Model → Response

API:

Your App → Internet → AI Provider → Response

4. If Ollama is installed correctly, why can a large model still run slowly?

Ollama doesn't remove hardware requirements—the model still needs enough compute and memory.

A large model can be slow because of:

- Limited RAM
- Weak CPU/GPU
- Insufficient VRAM
- Large model size
- Higher-precision model
- CPU-only inference
- Memory pressure causing swapping

For example, a small quantised model may run comfortably on a laptop, while a much larger model may generate tokens slowly.

5. Can Ollama always work without an internet connection?

No—not always.

Ollama can generally run models offline after the required model has already been downloaded.

First time:

Internet → Download model → Store locally → Run

Later:

Local model → Ollama → Generate response

Date: 7th Aug 2026

Topic: Vector Database

1. What is a Vector Database?

A Vector Database is a database designed to store, index, and search vector embeddings efficiently.

Examples: Pinecone, FAISS, Chroma, Weaviate, Milvus.

2. Why is a Vector Database used in Generative AI?

It is mainly used to help AI applications find relevant information quickly based on semantic meaning.

For example, if a document contains:

“Employees can take 20 paid vacation days.”

A user asks:

“How many annual leaves can an employee take?”

A vector search can understand that these two statements are related even though the exact words are different.

3. What is a Vector in GenAI?

A vector is a numerical representation of data such as text, images, or other content.

In GenAI, text is converted into a vector called an embedding. The embedding captures meaningful patterns or semantic information about the text.

Example:

"How can I apply for leave?"



Embedding Model



[0.21, -0.45, 0.78, ...]

4. What is the Difference Between a Normal Database and a Vector Database?

Normal Database	Vector Database
Stores structured/unstructured data	Primarily stores and indexes embeddings
Searches using values/conditions	Searches using vector similarity
Example: <code>WHERE employee_id = 101</code>	Finds semantically similar content
Good for exact data retrieval	Good for semantic similarity search
SQL databases are common	Vector indexes/databases are common

5. How Does a Vector Database Help a RAG Application?

A Vector Database is commonly used as the knowledge retrieval layer in RAG.

RAG Flow:

Documents



Split into chunks



Create embeddings



Store embeddings in Vector DB
↓
User asks a question
↓
Question → Embedding
↓
Vector DB finds relevant chunks
↓
Retrieved context + Question → LLM
↓
Final Answer

Regards
Ashka Singh

On Thu, 24 Sept 2026 at 20:16, ashka singh <sgsinghashka@gmail.com> wrote:

Hi Sir,

Regarding the topics of Components of Power BI, Power BI – Difference Between Dashboard and Report, Power BI – DAX (Data Analysis Expressions), Power BI – DAX Exercises, Deploying a Streamlit App on Streamlit Cloud, SQL CASE Statement, SQL Index, SQL Query Optimization, SQL Stored Procedure, SQL Trigger, SQL Window Functions. Kindly find the requested details below.

I'm submitting my pending assignments and projects as my final submission. Due to other ongoing commitments, I was unable to complete them within the scheduled timeline. I have already informed the Top Mentor Support Team regarding the delay.

Date: 13th June 2026

Topic: Components of Power BI

1. What is Power BI Desktop?

Power BI Desktop is a Microsoft tool used to connect data, clean it, analyze it, and create interactive reports and dashboards.

It is mainly used for data preparation, modeling, DAX calculations, and report creation.

2. What is the role of Power Query?

Power Query is used to clean and transform raw data before using it for analysis.

For example, we can remove duplicates, handle missing values, change data types, filter data, and merge tables.

3. What is DAX in Power BI?

DAX (Data Analysis Expressions) is the formula language used in Power BI to create measures, calculated columns, and calculated tables.

Example:

Total Sales = SUM(Sales[Amount])

4. What is the difference between Power Query and Power Pivot?

Power Query	Power Pivot
Used for data cleaning & transformation	Used for data modeling & calculations
Uses M language	Uses DAX
Works before data enters the model	Works on the data model
Remove duplicates, merge, append, filter	Relationships, measures, calculated columns

5. What are Visualizations in Power BI?

Visualizations are graphical ways of presenting data so that users can easily understand trends, comparisons, and KPIs.

Examples:

- Bar/Column Chart
- Line Chart
- Pie/Donut Chart
- Table/Matrix
- Card
- Map
- KPI

Date: 13th June 2026

Topic: Power BI – Difference Between Dashboard and Report

1. What is the difference between a Dashboard and a Report in Power BI?

Dashboard	Report
Usually a single-page view	Can have multiple pages
Shows high-level KPIs and important insights	Provides detailed analysis
Mainly available in Power BI Service	Created mainly in Power BI Desktop
Used for quick monitoring	Used for deeper analysis

2. Why are dashboards generally limited to a single page?

Dashboards are designed to give users a quick overview of important KPIs and business performance in one place.

Example: A manager can see Sales, Profit, Orders, and Revenue on one screen without going through multiple pages.

3. What are slicers and filters used for in reports?

Both are used to control and narrow down the data displayed in visuals.

- Slicer: A visual filter that users can directly click.
- Filter: Filters data at different levels, such as visual, page, or entire report.

Example: Select 2026 in a Year slicer to see only 2026 sales.

4. Which users typically prefer dashboards over reports?

Users who need quick, high-level information usually prefer dashboards.

Examples:

- Managers
 - Business leaders
 - Senior executives
 - Decision-makers
-

5. Can a dashboard be created using multiple reports in Power BI?

Yes. In Power BI Service, a dashboard can contain tiles from multiple reports and datasets.

For example, one dashboard can show:

- Sales KPIs from a Sales report
- Customer metrics from a Customer report
- Finance KPIs from a Finance report

Date: 13th June 2026

Topic: Power BI – DAX (Data Analysis Expressions)

1. What is DAX in Power BI?

DAX (Data Analysis Expressions) is the formula language used in Power BI for creating measures, calculated columns, and calculated tables.

Example:

Total Sales = SUM(Sales[Amount])

2. What is a Measure in Power BI?

A Measure is a DAX calculation that is calculated dynamically based on the current filter or visual context.

Example:

Total Sales = SUM(Sales[Amount])

If I select 2025, the measure automatically shows sales only for 2025.

3. Explain the purpose of **CALCULATE()**.

CALCULATE() is one of the most important DAX functions. It evaluates an expression after modifying the filter context.

Example:

Online Sales =

CALCULATE(

[Total Sales],

Sales[Channel] = "Online"

)

Here, **CALCULATE()** calculates Total Sales only for Online sales.

4. What is the difference between **COUNT()** and **COUNTROWS()**?

COUNT()**COUNTROWS()**

Counts non-blank values in a column

Counts rows in a table

Works on a specific column

Works on a table

Example:

Customer Count = **COUNT**(Customer[CustomerID])

Order Count = **COUNTROWS**(Orders)

5. How does Power BI execute a DAX formula containing multiple filters?

Power BI evaluates the DAX expression by applying the filter context and then calculating the result based on the filtered data.

For example:

Sales India =

CALCULATE(

[Total Sales],

Sales[Country] = "India",

Sales[Year] = 2026

)

Power BI effectively considers:

Total Sales → Country = India → Year = 2026 → Calculate result

If multiple filters are provided, they normally work together as AND conditions.

Date: 13th June 2026

Topic: Power BI – DAX Exercises

1. What is the difference between `COUNT()` and `COUNTROWS()` in DAX?

- `COUNT()` → Counts non-blank values in a column.
- `COUNTROWS()` → Counts the number of rows in a table.

Example:

Total Customers = `COUNT(Customer[CustomerID])`

Total Orders = `COUNTROWS(Orders)`

2. Explain how `CALCULATE()` works.

`CALCULATE()` evaluates an expression after changing or adding filters to the current filter context.

Example:

India Sales =

```
CALCULATE(  
    [Total Sales],  
    Sales[Country] = "India"  
)
```

Here, Power BI calculates Total Sales only for India.

3. What is the purpose of the `ALL()` function?

`ALL()` is used to remove filters from a table or column.

Example:

Total Sales All Countries =

```
CALCULATE(  
    [Total Sales],  
    ALL(Sales[Country])  
)
```

Even if the report is filtered to India, this measure calculates sales for all countries.

4. How does **RANKX()** create rankings?

RANKX() compares a value against a set of values and assigns a rank.

Example:

Product Rank =

```
RANKX(
    ALL(Product[ProductName]),
    [Total Sales],
    ,
    DESC
)
```

If Product A has the highest sales, it gets Rank 1.

5. Why do we use calculated columns and measures in Power BI?

Both are used for calculations, but they serve different purposes:

Calculated Column	Measure
Calculated row by row	Calculated based on filter context
Stored in the data model	Calculated when the visual/query needs it
Useful for categories, flags, row-level logic	Best for KPIs, totals, percentages
Increases model size	Usually more efficient for aggregations

Example:

Calculated Column:

Profit = Sales[Revenue] - Sales[Cost]

Measure:

Total Profit = SUM(Sales[Profit])

Date: 14th June 2026

Topic: Deploying a Streamlit App on Streamlit Cloud

1. What is deployment in web applications?

Deployment means making an application available on the internet/server so users can access and use it.

For example, after developing a Streamlit application locally, deploying it allows users to open it through a web URL without running Python on their own computer.

2. Why is GitHub required for Streamlit Cloud deployment?

GitHub is commonly used to store and manage the Streamlit application's source code.

Streamlit Cloud connects to the GitHub repository and uses the code from there to build and run the application.

3. What is the purpose of `requirements.txt`?

`requirements.txt` contains the Python libraries and their versions required by the application.

Example:

`streamlit`

`pandas`

`scikit-learn`

`matplotlib`

During deployment, Streamlit Cloud uses this file to install the required dependencies.

4. How does Streamlit Cloud update deployed applications?

When the connected GitHub repository is updated, Streamlit Cloud can detect the changes and redeploy the application.

For example:

Modify code → Git commit/push → Streamlit Cloud detects change → Rebuilds → Updated app

5. Explain the complete deployment workflow of a Streamlit application.

A simple workflow is:

1. Develop application locally
- ↓
2. Test the Streamlit app
- ↓
3. Create `requirements.txt`
- ↓
4. Push project to GitHub
- ↓
5. Connect GitHub repository to Streamlit Cloud
- ↓
6. Select the main Streamlit file, e.g. `app.py`
- ↓
7. Streamlit Cloud installs dependencies
- ↓
8. Application is built and deployed
- ↓
9. Get a public URL and share the application

Date: 20th June 2026

Topic: SQL CASE Statement

1. What is CASE Statement in SQL?

CASE is used to apply conditional logic in SQL, similar to IF-ELSE in programming.

It checks conditions and returns a value based on which condition is true.

Example:

```
SELECT Name,  
  
       CASE  
  
         WHEN Salary >= 50000 THEN 'High'  
  
         WHEN Salary >= 30000 THEN 'Medium'  
  
         ELSE 'Low'  
  
       END AS Salary_Category  
  
FROM Employees;
```

2. Why do we use CASE Statement?

We use **CASE** to create conditional results directly in SQL.

Common uses:

- Categorize data
- Create labels/groups
- Apply business rules
- Handle conditional calculations

Example:

Salary → High / Medium / Low

3. What is the syntax of CASE Statement?

CASE

WHEN condition1 THEN result1

WHEN condition2 THEN result2

WHEN condition3 THEN result3

ELSE result

END

4. What is the use of ELSE in CASE Statement?

ELSE provides a default value when none of the **WHEN** conditions are true.

Example:

CASE

WHEN Marks >= 60 THEN 'Pass'

ELSE 'Fail'

END

Here, students with marks below 60 get Fail.

5. Tricky: What happens if no WHEN condition is true and ELSE is not used?

If no WHEN condition is true and there is no ELSE, the CASE expression returns NULL.

Example:

CASE

 WHEN Marks >= 60 THEN 'Pass'

END

If Marks = 40 → Result is NULL.

Date: 20th June 2026

Topic: SQL Index

1. What is an Index in SQL?

An Index is a database structure that helps SQL find rows faster, similar to an index in a book.

For example, if we frequently search customers by CustomerID, creating an index on that column can speed up those searches.

2. Why do we use Index?

We use indexes mainly to improve query performance, especially for columns frequently used in:

- WHERE
- JOIN
- ORDER BY
- GROUP BY

Example:

If a table has millions of customers and we search by CustomerID, an index can make the lookup much faster.

3. What is the syntax to create an Index?

Basic syntax:

CREATE INDEX index_name

ON table_name (column_name);

Example:

```
CREATE INDEX idx_customer_id
```

```
ON Customers (CustomerID);
```

This creates an index on **CustomerID**.

4. What is a Composite Index?

A Composite Index is an index created on multiple columns.

Example:

```
CREATE INDEX idx_customer_city
```

```
ON Customers (CustomerID, City);
```

This can be useful when queries commonly filter or search using these columns together.

Single-column index = 1 column

Composite index = Multiple columns

5. Tricky: Why should we not create an Index on every column?

Because indexes are not free.

Every index:

- Takes additional storage
- Can slow down **INSERT**, **UPDATE**, and **DELETE** because the index also needs to be updated
- Adds maintenance overhead
- May provide little benefit on columns that aren't commonly searched

Date: 20th June 2026

Topic: SQL Query Optimization

1. What is Query Optimization in SQL?

Query Optimization means improving a SQL query so that it runs faster and uses fewer system resources while producing the same result.

Common techniques include:

- Using proper indexes
 - Avoiding unnecessary columns
 - Optimizing joins
 - Filtering data early
 - Checking the execution plan
-

2. Why should we avoid **SELECT ***?

SELECT * retrieves all columns, even when we don't need them.

Example:



SELECT *

FROM Employees;



SELECT EmployeeID, Name, Salary

FROM Employees;

Benefits:

- Less data transferred
- Lower memory usage
- Faster queries, especially with wide/large tables
- Makes the query clearer

3. How does Index help in Query Optimization?

An index helps the database find required rows more efficiently instead of scanning the entire table in many cases.

Example:

SELECT *

FROM Employees

WHERE EmployeeID = 101;

If **EmployeeID** is properly indexed, the database may use that index to locate the required row faster.

4. What is the use of **EXPLAIN** in SQL?

EXPLAIN shows how the database plans to execute a query.

It can help us identify:

- Whether indexes are being used
- Table scans
- Join methods
- Estimated rows
- Potential performance issues

Example:

EXPLAIN

SELECT *

FROM Employees

WHERE EmployeeID = 101;

5. Tricky: Can too many indexes slow down a database?

Yes. Too many indexes can slow down **INSERT**, **UPDATE**, and **DELETE** operations because the database has to maintain those indexes as data changes.

They also consume additional storage.

Date: 20th June 2026

Topic: SQL Stored Procedure

1. What is a Stored Procedure in SQL?

A Stored Procedure is a predefined set of SQL statements stored inside the database that can be executed whenever needed.

Example:

CREATE PROCEDURE GetEmployees

AS

SELECT * FROM Employees;

Stored Procedure = Reusable SQL code stored in the database

2. Why do we use Stored Procedures?

We use Stored Procedures to:

- Reuse SQL logic
 - Reduce repetitive code
 - Improve maintainability
 - Apply business logic at the database level
 - Control access to certain database operations
-

3. How do we execute a Stored Procedure?

1. What is a Stored Procedure in SQL?

A Stored Procedure is a predefined set of SQL statements stored inside the database that can be executed whenever needed.

Example:

```
CREATE PROCEDURE GetEmployees
```

```
AS
```

```
SELECT * FROM Employees;
```

Stored Procedure = Reusable SQL code stored in the database

4. What is the use of parameters in a Stored Procedure?

Parameters allow us to pass input values to a Stored Procedure, making it reusable for different situations.

Example:

```
CREATE PROCEDURE GetEmployeeByID
```

```
    @EmployeeID INT
```

```
AS
```

```
SELECT *
```

```
FROM Employees
```

```
WHERE EmployeeID = @EmployeeID;
```

Now we can execute:

```
EXEC GetEmployeeByID @EmployeeID = 101;
```

5. Tricky: Difference between Stored Procedure and Normal SQL Query?

Stored Procedure	Normal SQL Query
Stored in the database	Usually written/executed when needed
Can contain multiple SQL statements and logic	Usually performs a specific operation

Can accept parameters

Can also use parameters depending on the application/tool

Reusable

May need to be written again

Can include conditions, loops, error handling, etc.

Generally simpler

Date: 20th June 2026

Topic: SQL Trigger

1. What is a Trigger in SQL?

A Trigger is a database object that automatically executes when a specific event occurs on a table, such as INSERT, UPDATE, or DELETE.

Trigger = Automatic action when a database event happens

2. Why do we use Triggers?

Triggers are used to automatically perform actions when data changes.

Common uses:

- Maintain audit/history tables
- Validate data
- Automatically update related data
- Track changes

Example:

When an employee's salary is updated, a trigger can automatically store the old and new salary in an audit table.

3. What is the difference between BEFORE Trigger and AFTER Trigger?

BEFORE Trigger

AFTER Trigger

Runs before the database operation

Runs after the operation

Can be used to validate/modify data before it is saved

Useful for logging or actions after successful changes

Example: validate input before INSERT

Example: create an audit record after UPDATE

BEFORE = Before data change

AFTER = After data change

Note: Trigger syntax and supported timing vary by database system. For example, SQL Server uses **AFTER** and **INSTEAD OF** triggers rather than **BEFORE** triggers.

4. What is the use of **OLD** and **NEW** in Trigger?

OLD and **NEW** are used to access previous and new values during a data change.

- **OLD** → Value before the operation
- **NEW** → Value after/new value

Example during an update:

OLD Salary = 40,000

NEW Salary = 50,000

This is useful for auditing and tracking changes.

5. Tricky: Can we use **OLD** and **NEW** both in every Trigger?

No. Their availability depends on the database system and the type of operation.

In databases that support **OLD/NEW** row references, generally:

Operation	OLD	NEW
INSERT	✗	✓
UPDATE	✓	✓
DELETE	✓	✗

Date: 20th June 2026

Topic: SQL Window Functions

1. What is a Window Function in SQL?

A Window Function performs a calculation across a set of related rows without combining them into a single row.

Example:

```
SELECT Name, Salary,  
       AVG(Salary) OVER() AS Avg_Salary  
FROM Employees;
```

Window Function = Calculation across rows while keeping individual rows

2. Why do we use **OVER()** in SQL?

OVER() defines the window of rows on which the function should operate.

It can contain:

- **PARTITION BY** → divide rows into groups
- **ORDER BY** → define the order

Example:

```
SUM(Sales) OVER(ORDER BY SaleDate)
```

OVER() tells SQL which rows to consider.

3. Difference between **GROUP BY** and Window Functions?

GROUP BY	Window Function
Combines rows into groups	Keeps individual rows
Returns fewer rows	Usually returns same number of rows
Used for aggregation	Used for ranking, running totals, comparisons, etc.

Easy example:

GROUP BY → Department-wise total salary

Window function → Employee salary + department average salary on every employee row

4. Difference between **ROW_NUMBER()** and **RANK()**?

Suppose salaries are:

Employee	Salary
A	50,000
B	50,000
C	40,000

ROW_NUMBER():

1, 2, 3

RANK():

1, 1, 3

ROW_NUMBER() gives a unique sequence.

RANK() gives the same rank to ties and skips the next rank.

5. Difference between **RANK()** and **DENSE_RANK()**?

Using the same example:

Salary	RANK	DENSE_RANK
50,000	1	1
50,000	1	1
40,000	3	2

RANK() → skips rank after a tie

DENSE_RANK() → doesn't skip rank

6. What is the use of **PARTITION BY**?

PARTITION BY divides the data into separate groups for the window function.

Example:

```
RANK() OVER(  
  
    PARTITION BY Department  
  
    ORDER BY Salary DESC  
  
)
```

This ranks employees separately within each department.

PARTITION BY = Group data without collapsing rows

7. How can you find department-wise employee ranking?

```
SELECT Name, Department, Salary,
```

```
    RANK() OVER(  
  
        PARTITION BY Department  
  
        ORDER BY Salary DESC  
  
    ) AS Dept_Rank
```

```
FROM Employees;
```

This gives the highest-paid employee Rank 1 in each department.

8. What is the use of **LAG()**?

LAG() retrieves a value from a previous row.

Example:

```
SELECT SaleDate, Sales,
```

```
    LAG(Sales) OVER(ORDER BY SaleDate) AS Previous_Sales
```

```
FROM Sales;
```

Useful for:

- Month-to-month comparison
- Previous salary
- Previous transaction

LAG = Look backward

9. What is the use of **LEAD()**?

LEAD() retrieves a value from the following/next row.

SELECT SaleDate, Sales,

LEAD(Sales) OVER(ORDER BY SaleDate) AS Next_Sales

FROM Sales;

LEAD = Look forward

10. How can you calculate sales growth using SQL?

A common approach is to use **LAG()** to get previous-period sales.

SELECT Month, Sales,

LAG(Sales) OVER(ORDER BY Month) AS Previous_Sales,

(Sales - LAG(Sales) OVER(ORDER BY Month))

* 100.0 / LAG(Sales) OVER(ORDER BY Month) AS Growth_Percent

FROM MonthlySales;

Formula:

Growth % = (Current Sales – Previous Sales) / Previous Sales × 100

LAG() is commonly used for period-over-period growth.

11. What is the use of **NTILE()**?

NTILE(n) divides ordered rows into approximately equal-sized groups/buckets.

Example:

NTILE(4) OVER(ORDER BY Salary DESC)

This divides employees into 4 groups.

Common use:

- Quartiles
- Customer segmentation
- Top 25%, 50%, etc.

`NTILE(4)` → 4 buckets

12. What is the use of `CUME_DIST()`?

`CUME_DIST()` calculates the cumulative distribution of a row within the ordered dataset.

Its value ranges from 0 to 1.

For example, a value of `0.80` means approximately 80% of the rows are at or below that position, according to the ordering.

`CUME_DIST` = Relative position in the dataset

13. What is a Running Total?

A running total is the cumulative sum calculated from the beginning up to the current row.

Example:

Month	Sales	Running Total
Jan	100	100
Feb	200	300
Mar	150	450

Running Total = Previous Total + Current Value

14. How can you calculate a Running Total using a Window Function?

Use `SUM()` with `OVER()` and `ORDER BY`:

`SELECT SaleDate, Sales,`

`SUM(Sales) OVER(`

ORDER BY SaleDate

) AS Running_Total

FROM Sales;

For department-wise running totals:

SUM(Sales) OVER(

PARTITION BY Department

ORDER BY SaleDate

)

15. Why is **LAST_VALUE()** sometimes confusing?

The tricky part is the window frame.

For example:

LAST_VALUE(Salary) OVER(

ORDER BY Salary

)

may return the current row's value, rather than the overall last value, because the default window frame can end at the current row.

To get the actual last value across the complete window, explicitly define the frame:

LAST_VALUE(Salary) OVER(

ORDER BY Salary

ROWS BETWEEN UNBOUNDED PRECEDING

AND UNBOUNDED FOLLOWING

)

Regards
Ashka Singh

On Thu, 24 Sept 2026 at 19:06, ashka singh <sgsinghashka@gmail.com> wrote:

Hi Sir,

Regarding the topics of Machine Learning - Introduction_To_AI, Prompt Engineering, Prompt Template, Prompt Optimization, Components of Power BI, Connecting Data Sources in Power BI, Data Cleaning in Power BI, Data Refresh in Power BI, Introduction to Power BI, Power Query in Power BI, Power BI – Changing Colors in Visuals. Kindly find the requested details below.

I'm submitting my pending assignments and projects as my final submission. Due to other ongoing commitments, I was unable to complete them within the scheduled timeline. I have already informed the Top Mentor Support Team regarding the delay.

Date: 24th May 2026

Topic: Machine Learning - Introduction_To_AI

1. What is Artificial Intelligence and where is it used?

Artificial Intelligence is the ability of machines to perform tasks that normally require human intelligence, like learning, reasoning, and decision-making. It is used in areas like chatbots, recommendation systems, fraud detection, self-driving cars, and image recognition.

Keyword: AI → Machines performing intelligent tasks

2. Why is Data Preprocessing important in Machine Learning?

Data preprocessing is important because real-world data is usually messy. It may contain missing values, duplicates, outliers, or inconsistent formats. We clean and transform the data so the model can learn properly and give more reliable results.

Example: Handling missing values and scaling features before training KNN.

Keyword: Clean data → Better model performance

3. Difference between Label Encoding and One-Hot Encoding?

Label Encoding converts categories into numbers, like Male = 0 and Female = 1. One-Hot Encoding creates separate binary columns for each category. I generally prefer One-Hot Encoding when categories don't have any natural order.

Example:

Color = Red, Blue, Green

- **Label Encoding** → 0, 1, 2
- **One-Hot** → Red | Blue | Green → 1 | 0 | 0

4. Explain Bias and Variance with an example.

Bias means the model is too simple and misses important patterns, which leads to underfitting. **Variance** means the model is too sensitive to the training data, which can lead to overfitting.

Example:

- **High Bias** → Simple model performs poorly on both training and test data.
- **High Variance** → Model performs very well on training data but poorly on test data.

5. What is the role of MLOps in real-world AI projects?

MLOps helps us take a Machine Learning model from development to production and maintain it over time. It covers things like model deployment, monitoring, versioning, retraining, and automation.

Example:

"If customer behavior changes and the model's accuracy drops, MLOps helps us monitor that performance and retrain or redeploy the model."

keyword: Build → Deploy → Monitor → Retrain

Date: 30th May 2026

Topic: Prompt Engineering

1. What is Prompt Engineering?

Prompt Engineering is the process of designing and improving instructions given to an AI model to get accurate, relevant, and consistent responses.

Example: Instead of saying "Write an email," I can specify the role, purpose, tone, audience, and format to get a better result.

2. What is Zero-Shot Prompting?

Zero-shot prompting means asking an AI model to perform a task without providing any examples. We describe what we want the model to do.

Example:

"Classify this review as Positive or Negative: *The product is excellent.*"

No examples are given beforehand.

Key point: 0 examples → Zero-Shot

3. Difference between One-Shot and Few-Shot Prompting?

One-Shot prompting provides the model with one example before asking it to perform the task.
Few-Shot prompting provides multiple examples.

Example:

One-Shot:

"Good → Positive.
Amazing product → ?"

Few-Shot:

"Good → Positive
Terrible → Negative
Excellent → Positive
Average → ?"

Key point:

One-Shot = 1 example

Few-Shot = Multiple examples

4. What is Role-Based Prompting?

Role-based prompting means assigning a specific role or persona to the AI, so it responds according to that role and expertise.

Example:

"Act as a senior Python developer and review this code."

This gives the model a clear perspective and helps make the response more relevant.

5. Why is context important in a prompt?

Context is important because it tells the AI what the task is, who the audience is, and what constraints it should follow. Without enough context, the response can be generic or incorrect.

Example:

Instead of:

"Write a report."

We can say:

"Write a 500-word technical report for a management team explaining the benefits of implementing MLOps."

Date: 30th May 2026

Topic: Prompt Template

1. What is a Prompt Template?

A Prompt Template is a predefined structure for creating prompts, where we keep fixed instructions and use variables for changing information.

Example:

"Summarize the following {text} in {language}."

Here, {text} and {language} are variables.

2. Why are Prompt Templates important?

Prompt Templates are important because they make prompts consistent, reusable, and easier to maintain. Instead of writing the same prompt again and again, we can use one template with different inputs.

Key point: Reusable + Consistent + Easy to maintain

3. Explain the role field in Prompt Templates.

The role field defines how the AI should behave or respond. It gives the model a specific perspective or expertise.

Example:

Role: You are a senior data analyst.

Task: Analyze the given sales data.

4. Difference between a Good Prompt and a Bad Prompt?

Good Prompt	Bad Prompt
Clear and specific	Vague
Provides context	Little/no context
Defines the expected output	Doesn't specify format
Gives relevant constraints	No clear requirements

Example:

Bad:

"Write about Python."

Good:

"Explain Python to a beginner in 5 bullet points with simple examples."

5. How do Prompt Templates improve AI responses?

Prompt Templates improve responses by providing consistent instructions and context every time. This reduces ambiguity and makes the AI output more predictable and relevant.

Example:

For customer support, we can create one template that always specifies:

- Customer issue
- Professional tone

- Required response format
- Company policies

Date: 30th May 2026

Topic: Prompt Optimization

1. What is Prompt Optimization?

Prompt Optimization is the process of improving a prompt to get more accurate, relevant, and consistent responses from an AI model. We refine the instructions, context, examples, and output format based on the results.

2. Why is context important in prompts?

Context helps the AI understand the background and exact requirement of the task. Without proper context, the model may give a generic or irrelevant answer.

Example:

Instead of: *"Explain Python."*

Better: *"Explain Python to a beginner with 3 simple examples."*

3. Difference between Bad Prompt and Optimized Prompt?

A bad prompt is usually vague and lacks context, while an optimized prompt clearly defines the role, task, context, constraints, and expected output.

Bad Prompt:

"Write something about sales."

Optimized Prompt:

"Act as a sales analyst. Analyze the given monthly sales data and provide the top 3 trends in a simple table, followed by two business recommendations."

4. What are the key elements of an optimized prompt?

The main elements are:

- Role – Who should the AI act as?
- Task – What should it do?
- Context – What background does it need?

- Constraints – What rules or limitations should it follow?
 - Examples – Provide examples when required.
 - Output Format – How should the answer be structured?
-

5. Explain Prompt Optimization with an example.

Example - Initially, I use the prompt: 'Analyze this customer feedback.' The response may be too generic."

"I can optimize it by specifying the role, task, and output format: 'Act as a customer experience analyst. Analyze the feedback, identify the main complaints, classify the sentiment as Positive, Negative, or Neutral, and provide the result in a table.'"

"The optimized prompt gives the AI clearer instructions, so the output becomes more structured and useful."

Date: 7th June 2026

Topic: Components of Power BI

1. What is Power BI Desktop?

Power BI Desktop is a Microsoft tool used to connect, clean, analyze, and visualize data. We can create interactive reports and dashboards and then publish them to the Power BI Service.

Interview keywords: Connect → Transform → Analyze → Visualize

2. What is the role of Power Query?

Power Query is mainly used for data cleaning and transformation before analysis. We can remove duplicates, handle missing values, change data types, merge tables, and filter data.

Example:

If an Excel file has duplicate records and incorrect data types, I can clean them using Power Query.

3. What is DAX in Power BI?

DAX stands for Data Analysis Expressions. It is a formula language used to create measures, calculated columns, and calculated tables in Power BI.

Example:

Total Sales = SUM(Sales[Amount])

4. Difference between Power Query and Power Pivot?

The main difference is that Power Query is used for data preparation, while Power Pivot is used for data modeling and calculations using DAX.

Power Query	Power Pivot
Clean & transform data	Build data models
ETL process	Relationships
Merge/Append data	DAX calculations
M language	DAX language

5. What are Visualizations in Power BI?

Visualizations are graphical ways of representing data in Power BI, such as bar charts, line charts, pie charts, tables, cards, and maps. They help users understand trends, comparisons, and KPIs quickly.

Example:

A line chart can show monthly sales trends, while a card can show total revenue.

Date: 7th June 2026

Topic: Connecting Data Sources in Power BI

1. What is a Data Source in Power BI?

A data source is the place from where Power BI gets the data for analysis and reporting. It can be Excel, CSV, SQL Server, SharePoint, APIs, or cloud databases.

2. How do you connect an Excel file to Power BI?

In Power BI Desktop, I go to Get Data → Excel, select the Excel file, choose the required sheet or table, and click Load. If the data needs cleaning, I choose Transform Data to open Power Query.

3. What is the purpose of the Get Data option?

Get Data is used to connect Power BI with different data sources. It allows us to bring data from files, databases, cloud services, web sources, and other connectors into Power BI.

4. Difference between Import Mode and DirectQuery? (Tricky)

In Import Mode, data is loaded into Power BI's in-memory model, so reports are generally faster, but the data needs to be refreshed.

In DirectQuery, the data stays in the source, and Power BI sends queries to the source when users interact with the report.

Import	DirectQuery
Data loaded into Power BI	Data stays in source
Generally faster	Can be slower depending on source
Requires refresh for latest data	Queries source at report interaction
Good for most analytical workloads	Useful when data is very large or near-real-time access is needed

5. Which data sources have you worked with in Power BI?

I have mainly worked with Excel and CSV files for data analysis and reporting. I have used Power Query to clean and transform the data and then created relationships, DAX measures, and visualizations in Power BI.

Date: 7th June 2026

Topic: Data Cleaning in Power BI

1. What is Data Cleaning in Power BI?

Data cleaning is the process of finding and correcting issues in raw data before analysis. It includes handling missing values, removing duplicates, fixing data types, and correcting inconsistent values.

2. Why is Data Cleaning important before analysis?

Because poor-quality data can lead to wrong calculations and misleading insights. Cleaning ensures the data is accurate, consistent, and ready for analysis.

3. What is Power Query Editor?

Power Query Editor is the data preparation environment in Power BI. I use it to clean, transform, and reshape data before loading it into the Power BI data model.

Common tasks:

- Remove duplicates
 - Handle missing values
 - Change data types
 - Split/merge columns
 - Filter rows
 - Merge and append tables
-

4. How do you handle missing values in Power BI?

First, I check why the values are missing. Depending on the situation, I may remove those rows, replace the values with an appropriate value like 0 or 'Unknown', or keep them as null if the missing value has business meaning.

** I don't blindly replace missing values; I first understand the business context.**

5. What steps would you perform on the given CSV dataset before creating a dashboard?

First, I would load the CSV into Power BI and inspect the data. Then I would check data types, missing values, duplicates, incorrect or inconsistent values, and outliers. After cleaning, I would create relationships if there are multiple tables, create the required DAX measures, and validate the data before building the dashboard.

Date: 7th June 2026

Topic: Data Refresh in Power BI

1. What is Data Refresh in Power BI?

Data Refresh is the process of updating the data in a Power BI report or dataset from its original data source, so the report shows the latest available information.

Key point: Data Source → Refresh → Updated Report

2. Why is Data Refresh important?

It is important because business data keeps changing. Refresh ensures that the dashboard shows updated and accurate information instead of outdated data.

Example:

If daily sales are updated in Excel or SQL, refreshing the report brings those latest sales into Power BI.

3. How do you refresh data in Power BI Desktop?

In Power BI Desktop, I can click the Refresh button on the Home tab. Power BI reconnects to the data source, applies the Power Query transformations, and updates the data model.

Flow:

Home → Refresh → Data Source → Updated Model

4. Difference between Manual Refresh and Scheduled Refresh?

Manual Refresh is triggered by the user whenever they want to update the data. Scheduled Refresh automatically refreshes the dataset at predefined times in Power BI Service.

Manual Refresh	Scheduled Refresh
User triggers it	Happens automatically
Useful for on-demand updates	Useful for regular updates
Usually done in Desktop/Service	Configured in Power BI Service

5. Common reasons for refresh failure?

Some common reasons are incorrect or expired credentials, unavailable data sources, changed file paths, network issues, gateway problems, schema or column changes, and data type errors.

Date: 7th June 2026

Topic: Introduction to Power BI

1. What is Power BI?

Power BI is a Microsoft business intelligence and data visualization tool. It helps us connect to data, transform and analyze it, and create interactive reports and dashboards.

2. Why is Power BI used?

Power BI is used to turn raw data into meaningful business insights. It helps organizations track KPIs, identify trends, compare performance, and make data-driven decisions.

3. What are the main components of Power BI?

- Power BI Desktop → Build reports and data models
 - Power BI Service → Publish, share, and collaborate
 - Power BI Mobile → View reports on mobile devices
 - Power Query → Clean and transform data
 - Gateway → Connect Power BI Service to on-premises data sources
-

4. Difference between Dashboard and Report?

A report is usually a multi-page, detailed analysis with interactive visuals. A dashboard is generally a single-page view that highlights important KPIs and insights.

Report	Dashboard
Can have multiple pages	Single page
Detailed analysis	High-level overview
More interactive	Focuses on key metrics
Created mainly in Desktop	Created in Power BI Service

5. Name some data sources supported by Power BI.

Power BI supports many data sources such as Excel, CSV, SQL Server, SharePoint, web APIs, Azure, Oracle, and other cloud databases.

Date: 7th June 2026

Topic: Power Query in Power BI

1. What is Power Query in Power BI?

Power Query is the data transformation and preparation tool in Power BI. We use it to clean, transform, and prepare raw data before loading it into the data model.

Example: Removing duplicates, handling nulls, changing data types, filtering rows, etc.

2. Why is Power Query important?

Power Query is important because real-world data is rarely clean. It helps us prepare reliable and consistent data before analysis, which reduces errors in the final dashboard.

3. How do you remove duplicate records using Power Query?

In Power Query Editor, we select the column or columns that define the duplicate record, then go to Home → Remove Rows → Remove Duplicates. Power Query keeps one occurrence and removes the duplicates.

4. Difference between Merge Queries and Append Queries?

Merge combines columns from two tables based on a common key, similar to a SQL JOIN. Append combines rows from two or more tables with a similar structure.

Example:

- Merge: Customer table + Sales table → Add customer information to sales
 - Append: January Sales + February Sales → One combined Sales table
-

5. What does Close & Apply do in Power Query?

Close & Apply saves the transformations we've made in Power Query and loads the transformed data into the Power BI data model so we can use it for analysis and visualization.

Date: 7th June 2026

Topic: Power BI – Changing Colors in Visuals

1. How do you import a CSV file into Power BI?

In Power BI Desktop, we go to Home → Get Data → Text/CSV, select the CSV file, and preview the data. Then we either choose Load directly or Transform Data if the data needs cleaning.

Flow: **Get Data** → **Text/CSV** → **Select File** → **Load / Transform**

2. How can you change the color of a bar chart?

First, we select the bar chart, then go to the Format visual pane → Data colors, and choose the color we want.

3. What is the purpose of Data Colors in Power BI?

Data Colors allow us to control the colors of data points or categories in a visual. They help differentiate categories and make the chart easier to understand.

Example:

Sales → Blue, Expenses → Orange, Profit → Green.

4. How do you change the title color of a visual?

Select the visual, open the Format visual pane, go to General → Title or the visual's title settings, and change the Text/Color option.

5. Why should dashboards use consistent color themes?

Consistent colors make a dashboard easier to read and more professional. They also help users quickly understand the meaning of different categories across multiple visuals.

Regards
Ashka Singh

On Thu, 24 Sept 2026 at 18:01, ashka singh <sgsinghashka@gmail.com> wrote:

Hi Sir,

Regarding the topics of Algorithm Categories in Machine Learning, KNN in Machine Learning, Adding GST Project in Google Sheet, And() & Or() Operator in Google Sheet, And() & Or() Operator in Google Sheet, Filter() in Google Sheet, Freezing Row & Column in Google Sheet, Upper(), Lower(), Left(), Right(), Mid()

in Google Sheet, If() in Google Sheet, Google Sheets, Pivot Table in Google Sheet, Range Name of Cell in Google Sheet, sum() in google sheet, Sumif & Countif() in Google Sheet, Version History in Google Sheet, Vlookup() & Hlookup() in Google Sheet. Kindly find the requested details below.

I'm submitting my pending assignments and projects as my final submission. Due to other ongoing commitments, I was unable to complete them within the scheduled timelines. I have already informed the Top Mentor Support Team regarding the delay.

Algorithm Categories in Machine Learning

Date: 25th April 2026

1. What are the types of Machine Learning?

Mainly, Machine Learning is divided into three types: Supervised, Unsupervised, and Reinforcement Learning.

Supervised learning works with labeled data, unsupervised learning finds hidden patterns, and reinforcement learning learns through rewards and penalties.

Key point: Labeled data → Supervised | Patterns → Unsupervised | Rewards → Reinforcement

2. Difference between Regression and Classification?

The main difference is the type of output. Regression predicts a continuous numerical value, while classification predicts a category or class.

For example, predicting salary is regression, whereas predicting whether an employee will leave or stay is classification.

Key point: Number → Regression | Category → Classification

3. What is Clustering?

Clustering is an unsupervised learning technique where we group similar data points without predefined labels.

For example, an e-commerce company can group customers into low, medium, and high-spending segments.

Key point: No labels + Similar data → Groups

4. Give real-life examples of each type.

Supervised: Predicting house prices or detecting spam emails.

Unsupervised: Customer segmentation based on purchasing behavior.

Reinforcement: Training an AI to play games by giving rewards for good actions.

5. When should we use Classification?

We use classification when the output belongs to predefined categories.

For example, Spam/Not Spam, Pass/Fail, Fraud/Not Fraud, or Customer Churn/No Churn.

Date: 25th April 2026

Topic: KNN in Machine Learning

1. What is KNN algorithm?

KNN stands for K-Nearest Neighbors. It is a supervised learning algorithm that predicts the output of a data point based on the K closest data points in the dataset.

Example: If most of the nearest customers belong to the "High Spending" group, KNN can classify a new customer as High Spending.

2. How does KNN work step by step?

KNN works in a few simple steps:

1. Choose the value of K.
 2. Calculate the distance between the new data point and existing points.
 3. Select the K nearest points.
 4. For classification, take the majority class.
 5. For regression, take the average of their values.
-

3. What is the importance of K value?

K decides how many neighboring data points the algorithm considers for prediction. A small K can make the model sensitive to noise, while a large K can make it too generalized. So, we normally choose K using techniques like cross-validation.

Example:

K = 3 → considers 3 nearest neighbors

K = 10 → considers 10 nearest neighbors

4. What are the advantages and disadvantages of KNN?

Advantages:

- Simple and easy to understand
- No complex training process
- Works for both classification and regression

Disadvantages:

- Prediction can be slow for large datasets
 - Sensitive to irrelevant features and scaling
 - Choosing the right K is important
-

5. Why is KNN called Lazy Learning?

KNN is called lazy learning because it doesn't build a model during the training phase. It basically stores the training data and performs the actual computation when we make a prediction.

Easy way to remember:

Other algorithms → learn during training

KNN → learns at prediction time → Lazy Learning

Date: 26th April 2026

Topic: Adding GST Project in Google Sheet*

1. What is GST?

GST (Goods and Services Tax) is an indirect tax charged on the sale of goods and services in India.

Example: ₹1,000 product + 18% GST = ₹1,180 total.

2. How do you calculate GST in Google Sheets?

GST can be calculated in Google Sheets using the formula:

$\text{GST Amount} = \text{Price} \times \text{GST Rate} / 100$

For example, if the price is ₹1,000 and GST is 18%:

$=1000 \times 18\% \rightarrow ₹180$

$\text{Total Amount} = \text{Price} + \text{GST} = ₹1,180$

3. What is the formula for final price?

$\text{Final Price} = \text{Original Price} + \text{GST Amount}$

In Google Sheets:

$=\text{Price} + (\text{Price} \times \text{GST Rate} / 100)$

4. Where is GST used in real life?

GST is used in everyday transactions such as shopping, restaurants, online purchases, hotels, transportation, and other services. It is added to the price of taxable goods and services.

5. Why is GST project important?

A GST project is important because it helps understand tax calculation, data handling, and real-world business transactions. It also demonstrates practical skills in Google Sheets and basic data analysis.

Date: 26th April 2026

Topic: And() & Or() Operator in Google Sheet

1. What is AND() function?

The AND() function in Google Sheets checks whether multiple conditions are TRUE. It returns TRUE only when all conditions are satisfied; otherwise, it returns FALSE.

Example: =AND(A1>50,B1="Yes")

2. What is OR() function?

The OR() function in Google Sheets checks multiple conditions and returns TRUE if at least one condition is TRUE. It returns FALSE only when all conditions are FALSE.

Example: =OR(A1>50,B1="Yes")

3. Difference between AND() and OR()?

AND() returns TRUE only when all conditions are true, while OR() returns TRUE when at least one condition is true.

Example:

=AND(A1>50,B1="Yes") → Both conditions must be true.

=OR(A1>50,B1="Yes") → Either condition can be true.

4. How are they used with IF()?

AND() and OR() are used inside IF() to apply multiple conditions.

Example:

=IF(AND(A1>=50,B1="Yes"),"Eligible","Not Eligible")

This returns "Eligible" only when both conditions are sat

5. Give real-life example of AND() and OR().

AND() Example: A student is eligible for a scholarship if marks are above 80% AND attendance is above 75%.

OR() Example: A customer gets a discount if they are a member OR their purchase is above ₹5,000.

Date: 26th April 2026

Topic: Dragging in Google Sheet

1. What is dragging in Google Sheets?

Dragging in Google Sheets means copying a formula, value, or pattern to nearby cells by dragging the small blue square (fill handle) at the bottom-right of a selected cell.

2. What is fill handle?

The fill handle is the small blue square at the bottom-right corner of a selected cell in Google Sheets. It is used to copy formulas, values, or continue a series into nearby cells.

3. How does Google Sheets detect a pattern?

Google Sheets detects a pattern by analyzing the values or formulas in the selected cells and identifying a sequence or relationship.

Example: If you enter 1, 2, 3, Sheets can continue the pattern as 4, 5, 6 when dragged.

4. Give examples of dragging uses.

Dragging in Google Sheets can be used to:

- Copy formulas to multiple cells.
- Continue number sequences such as 1, 2, 3, 4.
- Continue dates such as Monday, Tuesday, Wednesday.
- Repeat values or patterns across rows or columns.

5. What happens if only one value is selected?

If only one value is selected, dragging the fill handle usually copies or repeats that value into the selected cells.

Date: 26th April 2026

Topic: Filter() in Google Sheet

1. What is FILTER() function?

The FILTER() function returns only the rows or columns that meet specified conditions.

Example: =FILTER(A2:C10,C2:C10="Yes")

2. What is syntax of FILTER()?

The syntax of FILTER() is:

=FILTER(range, condition1, [condition2, ...])

It returns only the data that matches the given conditions.

3. Does FILTER() change original data?

No. FILTER() does not change the original data. It creates a filtered result in a separate location.

4. Can we apply multiple conditions?

Yes. FILTER() can apply multiple conditions using * for AND logic.

Example: =FILTER(A2:C10,(B2:B10="Yes")*(C2:C10>50))

5. Give real-life example of FILTER().

Example: In a sales report, FILTER() can be used to display only sales above ₹50,000 or transactions from a specific region.

Date: 26th April 2026

Topic: Freezing Row & Column in Google Sheet

1. What is freezing in Google Sheets?

Freezing in Google Sheets keeps selected rows or columns visible while scrolling through the sheet.

2. How to freeze a row?

To freeze a row in Google Sheets:

View → Freeze → Select the number of rows you want to freeze.

3. How to freeze a column?

To freeze a column in Google Sheets:

View → Freeze → Select the number of columns you want to freeze.

4. Why is freezing useful?

Freezing keeps important headers or labels visible while scrolling, making large datasets easier to read and analyze.

5. Can we freeze both row and column together?

Yes. Google Sheets allows you to freeze both rows and columns at the same time to keep headers and labels visible while scrolling.

Date: 26th April 2026

Topic: average(), count(), min(), max() in google sheet

1. What is AVERAGE() function?

The AVERAGE() function calculates the average (mean) of a set of numbers.

Example: =AVERAGE(A2:A10)

2. What is the difference between COUNT() and SUM()?

COUNT() counts the number of cells containing numbers, while SUM() calculates the total of those numbers.

Example: =COUNT(A2:A10) → Number of entries

=SUM(A2:A10) → Total value

3. How does MIN() work?

The MIN() function returns the smallest value from a range of numbers.

Example: =MIN(A2:A10) → Returns the lowest value.

4. Where is MAX() used?

The MAX() function is used to find the largest value in a range of numbers.

Example: =MAX(A2:A10) → Returns the highest value.

5. Can COUNT() count text values?

No. COUNT() only counts numeric values. It does not count text or blank cells.

Example: =COUNT(A2:A10)

Date: 26th April 2026

Topic: Upper(), Lower(), Left(), Right(), Mid() in Google Sheet

1. What is UPPER() function?

The UPPER() function converts text into uppercase letters.

Example: =UPPER("hello world") → HELLO WORLD

2. What is difference between LEFT() and RIGHT()?

LEFT() extracts characters from the beginning (left side) of a text, while RIGHT() extracts characters from the end (right side).

Example: =LEFT("Google",3) → Goo

=RIGHT("Google",3) → gle

3. How does MID() work?

The MID() function extracts a specific number of characters from the middle of a text, starting from a given position.

Example: =MID("Google",2,3) → oog

4. Where are text functions used?

Text functions are used to clean, format, and extract information from text data.

Examples: Formatting names, extracting IDs, splitting addresses, and standardizing data.

5. Why is LOWER() useful?

The LOWER() function converts text into lowercase letters. It is useful for standardizing text data and maintaining consistent formatting.

Example: =LOWER("HELLO WORLD") → hello world

Date: 26th April 2026

Topic: If() in Google Sheet*

1. What is IF() function?

The IF() function checks a condition and returns one value if TRUE and another value if FALSE.

Example: =IF(A2>=50,"Pass","Fail")

2. What is syntax of IF()?

The syntax of IF() is:

=IF(condition, value_if_true, value_if_false)

It checks a condition and returns the appropriate result.

3. Give real-life example of IF().

Example: In a student report, IF() can determine the result based on marks.

=IF(A2>=40,"Pass","Fail")

If marks are 40 or above → Pass, otherwise → Fail.

4. Can IF() be combined with other functions?

Yes. IF() can be combined with functions like AND(), OR(), SUM(), and COUNT() to apply more complex conditions.

Example: =IF(AND(A2>=50,B2="Yes"),"Eligible","Not Eligible")

5. What happens if condition is false?

If the condition is FALSE, the IF() function returns the value specified for the false condition.

Example: =IF(A2>=40,"Pass","Fail") → If A2 is below 40, it returns "Fail".

Date: 26th April 2026

Topic: Google Sheets

1. What is Google Sheets?

Google Sheets is a cloud-based spreadsheet application used to organize, analyze, calculate, and visualize data online.

2. How is Google Sheets different from Excel?

Google Sheets is mainly cloud-based and supports real-time collaboration, while Excel is a desktop-based spreadsheet application with more advanced data analysis features.

Both are used to organize, calculate, and analyze data.

3. What is a cell and how is it identified?

A cell is the basic unit in a spreadsheet where data is entered. It is identified by its column letter and row number.

Example: A1 means column A, row 1.

4. What are the advantages of using Google Sheets?

The main advantages of Google Sheets are:

- Cloud-based and accessible from anywhere.
- Real-time collaboration with multiple users.
- Automatic saving of changes.
- Useful formulas, functions, charts, and data analysis tools.

5. How can multiple users work on the same sheet?

Multiple users can work on the same Google Sheet by sharing it with their email addresses and giving them appropriate access, such as Viewer, Commenter, or Editor. Changes are updated in real time.

Date: 26th April 2026

Topic: Pivot Table in Google Sheet

1. What is a pivot table?

A Pivot Table is a tool used to summarize, organize, and analyze large amounts of data by grouping information and calculating totals, counts, or averages.

2. How do you create a pivot table?

To create a Pivot Table in Google Sheets:

Select the data → Insert → Pivot table → Choose the data range and location → Click Create.

3. Does pivot table change original data?

No. A Pivot Table does not change the original data. It creates a separate summary of the data for analysis.

4. What are rows and values in pivot table?

Rows group data into categories, while Values perform calculations such as SUM, COUNT, or AVERAGE on the data.

Example: Rows = Product, Values = Total Sale

5. Give real-life example of pivot table.

A sales team can use a Pivot Table to summarize total sales by product, region, or month and quickly identify sales trends.

Date: 26th April 2026

Topic: Range Name of Cell in Google Sheet

1. What is a range name in Google Sheets?

A range name is a custom name assigned to a cell or range of cells in Google Sheets, making it easier to reference them in formulas.

Example: Naming A2:A10 as SalesData.

2. How do you create a named range?

Select the cells → Data → Named ranges → Enter a name → Done.

3. What are the rules for naming range?

Rules for naming a range:

- The name must start with a letter.
- It can contain letters, numbers, and underscores (_).
- It cannot contain spaces.
- It cannot be the same as a cell reference, such as A1.

4. Why are range names useful?

Range names make formulas easier to read, understand, and manage by replacing cell references with meaningful names.

Example: =SUM(SalesData) is easier to understand than =SUM(A2:A100).

5. Can we edit a named range?

Yes. We can edit a named range by going to Data → Named ranges, then changing its name or cell range.

Date: 26th April 2026

Topic: sum() in google sheet

1. What is SUM() function?

The SUM() function adds numbers together and returns their total.

Example: =SUM(A2:A10) → Calculates the total of the values.

2. What is the syntax of SUM()?

The syntax of SUM() is:

=SUM(value1, [value2, ...])

Example: =SUM(A2:A10) → Adds all values from A2 to A10.

3. Where is SUM() used in real life?

SUM() is used to calculate total sales, expenses, revenue, marks, salaries, or monthly budgets.

Example: =SUM(B2:B10) → Calculates total sales.

4. Can SUM() ignore text values?

Yes. SUM() ignores text values and adds only the numeric values in the selected range.

Example: =SUM(A2:A10)

5. What is the difference between range and individual sum?

A range adds all values within a continuous range, while an individual SUM adds specific cells.

Example:

=SUM(A2:A10) → Adds all cells from A2 to A10.

=SUM(A2,A5,A8) → Adds only A2, A5, and A8.

Date: 26th April 2026

Topic: Sumif & Countif() in Google Sheet

1. What is SUMIF()?

SUMIF() adds values that meet a specific condition.

Example: =SUMIF(A2:A10,"Apple",B2:B10) → Adds sales for Apple only.

2. What is COUNTIF()?

COUNTIF() counts the number of cells that meet a specific condition.

Example: =COUNTIF(A2:A10,"Yes") → Counts cells containing "Yes".

3. What is the difference between SUM and SUMIF()?

SUM() adds all numeric values, while SUMIF() adds only the values that meet a specified condition.

Example:

=SUM(B2:B10) → Adds all values.

=SUMIF(A2:A10,"Apple",B2:B10) → Adds only Apple sales.

4. Give a real-life example of COUNTIF().

Example: In a sales report, COUNTIF() can count how many orders have the status "Completed".

=COUNTIF(C2:C100,"Completed")

5. What type of conditions can be used?

Conditions can include text, numbers, comparisons, and logical criteria.

Examples: "Yes", ">50", "<100", "Apple", or "<>0".

Date: 26th April 2026

Topic: Version History in Google Sheet

1. What is version history in Google Sheets?

Version History in Google Sheets records changes made to a spreadsheet, allowing users to view or restore earlier versions.

2. How can you open version history?

Go to File → Version history → See version history to view previous versions of the Google Sheet.

3. Can we restore old data using version history?

Yes. Version History allows you to restore an earlier version of a Google Sheet if needed.

4. Why is version history important in teamwork?

Version History is important in teamwork because it tracks changes, shows who made them, and allows previous versions to be restored.

5. What information does version history show?

Version History shows who made changes, when they were made, and what changes were made to the spreadsheet.

Date: 26th April 2026

Topic: Vlookup() & Hlookup() in Google Sheet

1. What is VLOOKUP()?

VLOOKUP() searches for a value in the first column of a range and returns a related value from another column in the same row.

Example: =VLOOKUP(A2,D2:F10,3,FALSE)

2. What is difference between VLOOKUP() and HLOOKUP()?

VLOOKUP() searches vertically in the first column, while HLOOKUP() searches horizontally in the first row.

VLOOKUP = Vertical

HLOOKUP = Horizontal

3. What is search_key?

The search_key is the value you want to find in a VLOOKUP or HLOOKUP.

Example: In =VLOOKUP(A2,D2:F10,3,FALSE), A2 is the search_key.

4. Why do we use FALSE in lookup?

We use FALSE in VLOOKUP() to find an exact match for the search key.

Example: =VLOOKUP(A2,D2:F10,3,FALSE) → Searches for an exact match.

5. Give real-life example of lookup function.

Example: In a sales report, VLOOKUP() can find a product ID and return its product name or price from a product table.

=VLOOKUP(A2,D2:F10,2,FALSE)

Thank you
Regards
Ashka Singh

On Sat, 25 Apr 2026 at 18:01, ashka singh <sgsinghashka@gmail.com> wrote:

Dear Sir,

Regarding the topics of the Decision Trees, What is Kaggle?, Linear Regression, Logistic Regression, Random Forest, Feature Selection, LLM (Large Language Model), Model Score, One-Hot Encoding, Random State (42), Checking Shape of X_train, and Train-Test Split. Kindly find the requested details below.

Date: 18th April 2026
Topic: Decision Trees

1. What is decision tree?

Ans: A Decision Tree is a non-parametric supervised learning algorithm used for both classification (categorizing data) and regression (predicting numerical values).

2. What are parts of decision tree?

Ans: Decision trees in machine learning are structured as a tree of nodes and branches that map decisions, with main components including the root node (initial data), internal/decision nodes (feature tests), branches (outcomes), and leaf nodes (final classification/regression result).

3. Where is decision tree used?

Ans: Decision trees are used in machine learning for both [classification](#) (predicting categorical labels) and [regression](#) (predicting continuous values) tasks.

4. What is overfitting in decision trees?

Ans: Overfitting in decision trees occurs when the model grows too complex, memorizing the training data—including its noise and outliers—rather than learning general patterns.

5. What is pruning?

Ans: An optimization technique used to reduce model complexity by removing unnecessary components—such as branches in decision trees or weights in neural networks—to reduce overfitting and improve computational efficiency.

Date: 18th April 2026

Topic: What is Kaggle?

1. What is Kaggle?

Ans: Kaggle is an online platform for data science and machine learning.

-It provides datasets, notebooks, and competitions.

- Owned by Google.

2. What are Kaggle competitions?

Ans: A Kaggle competition is an online data science and machine learning challenge where participants build models to solve complex real-world problems, often for cash prizes and recognition.

3. How does Kaggle help beginners?

Ans: Kaggle helps beginners by providing free, hands-on data science resources, including over 500,000 public datasets, interactive courses, and free cloud-based GPU notebooks.

4. What is Kaggle notebook?

Ans: A Kaggle Notebook is a free, cloud-based Integrated Development Environment (IDE) that enables users to write, run, and share code (Python or R) directly in a browser.

5. Why is Kaggle important in ML?

Ans : Kaggle is crucial in machine learning (ML) as the premier platform for practical skill development, offering curated datasets, free GPU-powered notebooks, and community-driven learning.

Date: 18th April 2026

Topic: Linear Regression

1. What is linear regression?

Ans: Linear regression is a foundational supervised machine learning algorithm that models the linear relationship between a dependent (target) variable and one or more independent (feature) variables to predict continuous numerical outcomes.

2. Explain equation $y = mx + c$.

Ans: The equation $y = mx + c$ is the fundamental mathematical representation of Simple Linear Regression, a supervised machine learning algorithm used to model relationships between variables.

3. Where is linear regression used?

Ans: It is primarily applied for predictive modeling (forecasting), trend analysis, and feature importance evaluation across fields like finance, healthcare, and real estate.

4. What are limitations of linear regression?

Ans: Linear regression is limited by its strict assumption of a linear relationship between features, high sensitivity to outliers, and vulnerability to multicollinearity.

5. How does model learn best-fit line?

Ans: Models learn the best-fit line in machine learning—typically through [linear regression](#)—by minimizing the total error, or distance, between data points and a straight line ($y = mx + b$).

Date: 18th April 2026

Topic: Logistic Regression

1. What is logistic regression?

Ans: Logistic regression is a fundamental method used for binary classification (predicting 0 or 1, Yes/No) by modeling the probability of an event.

2. Why is it used for classification?

Ans: [Classification](#) is used to automatically categorize data into predefined classes (e.g., spam/not-spam, approved/rejected) by learning patterns from labeled training data.

3. What is sigmoid function?

Ans: The sigmoid function (specifically the logistic function) is a fundamental mathematical tool in machine learning used to map any real-valued number into a range between 0 and 1.

4. Difference between linear and logistic regression?

Ans: [Linear regression](#) predicts continuous numeric outcomes (e.g., prices, temperature) using a straight line, while [logistic regression](#) predicts categorical outcomes (e.g., Yes/No, Spam/Ham) by mapping output to probabilities between 0 and 1 using a sigmoid function.

5. Where is logistic regression used?

Ans: Logistic regression is a fundamental method used primarily for classification tasks, predicting the probability of an event (e.g., yes/no, 0/1) based on input data. It is widely used in finance for fraud detection, healthcare for disease diagnosis, and marketing for calculating the probability of a customer clicking an advertisement or churning.

Date: 18th April 2026

Topic: Random Forest

1. What is Random Forest?

Ans: Random Forest is a supervised learning algorithm.

- It is an ensemble method that combines multiple decision trees.
- Used for classification and regression.

2. Why is it called Random Forest?

Ans: Because it builds a "forest" (ensemble) of decision trees, where each tree is trained on a [random sample of data](#) (bootstrapping) and makes decisions based on a [random subset of features](#).

3. How does Random Forest reduce overfitting?

Ans: Random Forest reduces overfitting by combining multiple individual [decision trees](#) into an ensemble, which averages out the high variance and errors of single trees.

4. Where is Random Forest used?

Ans: Random Forest is a highly versatile [supervised machine learning algorithm](#) used across numerous industries to solve complex classification and regression problems.

5. Difference between Decision Tree and Random Forest?

Ans: A Decision Tree is a single, flowchart-like model that makes decisions based on a series of questions. A Random Forest is an ensemble method that combines the predictions of hundreds of individual decision trees to achieve a more accurate and stable result

Date: 19th April 2026

Topic: Feature Selection

1. What is feature selection?

Ans: Feature Selection means choosing **important input features (columns)**.

- It removes **unnecessary or irrelevant data**.
- Helps the model focus on **useful information**.

2. Why is it important?

Ans: Improves model performance

- Reduces overfitting
- Makes the model faster
- Simplifies the dataset

3. Difference between filter and wrapper methods?

Ans: Filter methods use statistical measures to score features independently of any machine learning model, while wrapper methods use a specific machine learning model to evaluate subsets of features by iteratively training and testing them.

4. What is embedded method?

Ans: Embedded methods are machine learning feature selection techniques that incorporate variable selection directly into the model training process.

5. Can too many features harm model?

Ans: Yes, too many features can significantly harm a machine learning model.

Date: 19th April 2026

Topic: LLM (Large Language Model)

1. What is LLM?

Ans: A type of AI, built on transformer neural networks, trained on massive datasets to understand and generate human-like text and code.

2. How does LLM work?

Ans: Based on **Transformer architecture**

- Learns patterns from **large datasets**.

- Predicts the next word/token in a sentence.

3. What is Transformer in LLM?

Ans: A deep learning neural network architecture that processes entire sequences of data simultaneously using self-attention mechanisms, rather than sequentially.

4. What are applications of LLM?

Ans: Chatbots & Virtual Assistants: Powering tools like ChatGPT, Claude, and Gemini for conversation and customer support.

- Content Generation: Producing articles, code, marketing copy, and creative writing.
- Search & Summarization: Organizing information, searching documents, and creating concise summaries
- Language Translation: Translating text between languages with high nuance.
- Sentiment Analysis: Evaluating customer feedback in e-commerce and finance.
- Coding Assistance: Generating, debugging, and explaining code in development tools.
- Information Retrieval: Answering complex queries by analyzing huge volumes of data.

5. What are limitations of LLM?

Ans: Hallucinations (confident falsehoods), static knowledge (outdated info), limited reasoning, bias, and high computational costs

Date: 19th April 2026

Topic: Model Score

1. What is model score?

Ans: Model Score tells how well a model **performs on data**.

- It measures **accuracy or performance** of the model.
- In regression, it is usually **R² Score** (coefficient of determination)

2. What is R² score?

Ans: A statistical metric in machine learning that measures the goodness of fit of a regression model by indicating the percentage of variation in the target variable explained by the independent variables.

3. How is model score calculated?

Ans: By comparing predicted values against actual, known outcomes (ground truth) using specific evaluation metrics.

4. Why use test data for scoring?

Ans: It provides an objective measure of how well a model will perform in the real world.

5. Can a high score be misleading?

Ans: Yes, a high score in machine learning can be, and often is, highly misleading.

Date: 19th April 2026

Topic: One Hot Encoding

1. What is one hot encoding?

Ans: One Hot Encoding converts **categorical data into binary (0/1) format**.

- Each category becomes a **separate column**.
- Only one column has **1**, others are **0**.

2. Why is it better than label encoding?

Ans: It avoids introducing false numerical relationships.

3. What is dummy variable trap?

Ans: A scenario in regression analysis where one-hot encoded categorical variables are highly correlated (perfect multicollinearity), causing model instability

4. When should we use one hot encoding?

Ans: One-hot encoding should be used for nominal (unordered) categorical data when working with algorithms that require numerical inputs, such as linear regression, logistic regression, or neural networks

5. What are its limitations?

Ans: Increases number of columns

- Can cause **high dimensionality**
- Not efficient for large categories

Date: 19th April 2026

Topic: Random State (42)

1. What is random_state?

Ans: Random state is a **parameter used to control randomness**.

- It ensures that results are **repeatable (reproducible)**.
- Used in functions like `train_test_split`, model training, etc.

2. Why do we use 42?

Ans: `42` is just a **number (seed value)**.

- It fixes the **random behavior**.
- You can use any number like `0`, `1`, `100`, etc.

3. Does random_state affect accuracy?

Ans: Yes, random_state directly affects the reported accuracy of a machine learning model, though it does not change the fundamental "power" of the underlying algorithm.

4. What happens if we don't use it?

Ans: If you don't use random_state (or set it to None), the code will produce different results every time you run it.

5. Can we change the seed value?

Ans: Yes, can and should change the seed value in random_state.

Date: 19th April 2026

Topic: Checking Shape of X_train

1. What does `.shape` return?

Ans: The `.shape` attribute returns a tuple of integers that indicates the number of elements along each dimension (axis) of a data structure.

2. Why check `X_train` shape?

Ans: - To confirm data is **correctly split**

- To know how many **training samples** are present

- To verify **number of features**

- Helps avoid errors during model training.

3. Difference between rows and columns?

Ans: Typically represented as (rows, columns) or (samples, features), it defines the structure of your training data. Rows represent individual data samples (observations), while columns represent the features (variables) associated with those samples.

4. What is feature in dataset?

Ans: `X_train` represents the training subset of your input features (independent variables) after a dataset has been split. The shape of `X_train`—usually accessed via `X_train.shape` in NumPy or pandas—is typically a 2D tuple: (number of samples, number of features).

5. What happens if shape is incorrect?

Ans: An incorrect shape leads to different consequences depending on the context, ranging from programming errors and manufacturing defects to geometric misconceptions or structural issues in design.

Date: 19th April 2026

Topic: Checking Shape of `X_train`

1. What is train test split?

Ans: Train-Test Split is a technique used to **divide a dataset into two parts**.

-One part is used to **train the model**.

-Second part is used to **test the model performance**.

2. Why is it important?

Ans: To check how model performs on **unseen data**.

- Helps to avoid **overfitting**.

- Gives an idea about **real-world performance**.

3. What is `test_size` parameter?

Ans: The `test_size` parameter in the [train_test_split](#) function from the scikit-learn library defines the proportion or absolute number of samples that should be allocated to the test dataset.

4. What is `random_state`?

Ans: A `random_state` in machine learning is a parameter (often integer) acting as a seed for a pseudo-random number generator, ensuring reproducibility.

5. What happens if we don't split data?

Ans: Not splitting data into training and testing sets usually causes [overfitting](#).

Thanks & Regard
Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>
Date: Sat, 18 Apr 2026 at 17:51
Subject: Fwd: TA || Ashka || TM- B141
To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

As mentioned in the last session, please find below the Blinkit project link:

<https://github.com/sgsinghashka-del/Blinkit-Dashboard/blob/main/README.md>

Regards
Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>
Date: Sat, 11 Apr 2026 at 16:28
Subject: Fwd: TA || Ashka || TM- B141
To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of the Basic UI Elements – title, header, text, Streamlit Buttons and Callbacks, Index Basics, INNER JOIN, Introduction to Views, LEFT JOIN, Streamlit Selection Widgets – Selectbox, Multiselect, Radio and Streamlit Input Widgets – text_input, number_input, kindly find the requested details below.

Date: 4th April 2026

Topic: Basic UI Elements – title, header, text
Technology: SQL

1. What is the difference between st.title() and st.header()?

Ans: The primary difference between st.title() and st.header() in Streamlit is their hierarchy and visual size. They are both wrappers around Markdown that allow us to display text in a structured way.

2. When should you use st.text() instead of st.write()?

Ans: In Streamlit, we should use st.text() when we need to display raw, preformatted text exactly as it is, without any styling or Markdown parsing. In contrast, st.write() is a multipurpose "Swiss Army knife" that attempts to intelligently format its input.

3. Can we use multiple titles in one app?

Ans: Yes, you can manage multiple titles for a single application in SQL.

4. How do UI elements improve user experience?

Ans: UI elements improve user experience by making digital interfaces intuitive, efficient, and engaging.

5. Which function is best for large headings?

Ans: `SELECT TOP 100 * FROM table_name;`

Date: 4th April 2026

Topic: Streamlit Buttons and Callbacks

Technology: SQL

1. What is the purpose of `st.button()` in Streamlit?

Ans: The primary purpose of `st.button()` in Streamlit is to create a clickable button widget that allows users to trigger specific actions or execute code blocks only when clicked.

2. What is a callback function?

Ans: A callback function is a function passed into another function as an argument, which is then executed (or "called back") inside the outer function to complete a specific routine, action, or asynchronous operation.

3. How does `session_state` help with buttons?

Ans: `st.session_state` in Streamlit helps with buttons by persisting data across the script's reruns, which normally reset whenever a button is clicked. Because Streamlit reruns the entire script from top to bottom on interaction, session state allows us to store the button's state ("pressed" vs "not pressed") or other data, preventing it from disappearing.

4. What happens if two buttons have the same callback?

Ans: When two buttons share the same callback, clicking either button triggers the exact same function.

5. How can you prevent a function from running repeatedly in Streamlit?

Ans: To prevent a specific function from running repeatedly during these reruns, we can use built-in tools like caching, session state, forms, and fragments.

Date: 4th April 2026

Topic: Index Basics Technology: SQL

1. What is an index in SQL?

Ans: Indexes in SQL are special database structures that speed up data retrieval by allowing quick access to records instead of scanning the entire table. They act like a lookup system and play an important role in improving query performance and database efficiency.

- It speed up queries (`SELECT`, `JOIN`, `WHERE`, `ORDER BY`).
- Reduce disk I/O and improve efficiency in large tables.
- Ensure data integrity with unique indexes.

2. How does an index improve performance?

Ans: Indexes improve SQL query performance by reducing the amount of data the database must scan to find results. Without an index, the database performs a full table scan, which checks every row in a table to satisfy a query. This is inefficient, especially for large tables.

3. Can index slow down operations?

Ans: Yes, database indexes can significantly slow down certain operations, specifically data modification, even though they are designed to speed up data retrieval (SELECT queries).

4. What is a composite index?

Ans: A composite index (also known as a multi-column index) is a database index created on two or more columns of a table.

5. When should we avoid using index?

Ans: Avoid using database indexes on small tables, columns with low cardinality (few unique values), or tables with frequent write operations (INSERT, UPDATE, DELETE).

Date: 4th April 2026

Topic: INNER JOIN

Technology: SQL

1. What is INNER JOIN?

Ans: An INNER JOIN in SQL selects records that have matching values in both tables, returning only the rows where the join condition is met.

2. When do we use INNER JOIN?

Ans: Use an INNER JOIN in SQL when we need to combine rows from two or more tables based on a related column, returning *only* the records that have matching values in both tables.

3. What happens if no matching record exists?

Ans: When no matching record exists in a SQL join, the database typically returns NULL values for all columns from the table that lacks a match. Specifically, with a LEFT JOIN, all rows from the left table are returned, and if no match exists on the right, the right-side columns display as NULL.

4. Difference between INNER JOIN and LEFT JOIN?

Ans:

Feature	INNER JOIN	LEFT (OUTER) JOIN
Result Set	Includes only rows with matching values in both tables.	Includes all rows from the left table and matched rows from the right.
Unmatched Rows	Discarded entirely.	Preserved from the left table; right-side columns are filled with NULL.
Mathematical Concept	Intersection ($\cap$).	Left table ($\cup$) + Intersection ($\cap$)

.)

Common Use Case	When you only want "complete" records (e.g., customers who have made orders).	When you want to see everything from one list, even if details are missing (e.g., all students, regardless of enrollment).
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5. What column is used for INNER JOIN?

Ans: An INNER JOIN uses a related column (often a foreign key in one table and the primary key in another) to match rows between two tables. The columns must share a common value or relationship, defined in the ON clause (e.g., TableA.id = TableB.foreign_id) to return only rows with matches in both tables.

Date: 4th April 2026

Topic: Introduction to Views

Technology: SQL

1. What is a view in SQL?

Ans: A view in SQL is a virtual table based on the result-set of an SQL statement.

2. Difference between table and view?

Ans: In a relational database, the fundamental difference is that a Table is a physical storage container for data, while a View is a "virtual table" that is actually a stored SQL query.

3. Why do we use views?

Ans: In SQL, a view is a virtual table based on the result-set of an SQL statement. Unlike a standard table, it does not store data physically but instead pulls data from underlying "base" tables dynamically when queried.

4. Can views improve security?

Ans: Yes, SQL views significantly improve security by restricting user access to sensitive data, limiting visibility to specific rows or columns, and enforcing read-only permissions. By granting users access to a view rather than the underlying base table, organizations can hide sensitive information, reduce data exposure risk, and enforce data integrity, as detailed by [OneUptime](#).

5. How do you delete a view?

Ans: In database management, "delete view" typically refers to permanently removing the view's definition from the database. This is performed using the DROP VIEW statement.

Date: 4th April 2026

Topic: LEFT JOIN

Technology: SQL

1. What is LEFT JOIN?

Ans: A LEFT JOIN (or LEFT OUTER JOIN) in SQL returns all records from the left table and matched records from the right table.

3. What does NULL indicate in LEFT JOIN result?

Ans: In a LEFT JOIN (or LEFT OUTER JOIN), a NULL result typically indicates that no matching record was found in the right table for a row existing in the left table.

4. When should we use LEFT JOIN?

Ans: Use a LEFT JOIN (or LEFT OUTER JOIN) when we want to retrieve all records from the left table and only the matching records from the right table.

5. Which table is called left table?

Ans: In SQL, the left table is the first table mentioned in the FROM clause or the table that appears immediately to the left of the [LEFT JOIN](#) keyword.

Date: 4th April 2026

Topic: Streamlit Selection Widgets – Selectbox, Multiselect, Radio

Technology: SQL

1. What is the difference between selectbox() and radio()?

Ans: [selectbox\(\)](#) (dropdown) and [radio\(\)](#) (radio buttons) both allow single-choice selection from a list but differ in visibility and space usage. Selectbox hides options until clicked, making it ideal for >6 -10 choices or limited space, while radio buttons reveal all choices, offering better visibility for 5 < Options.

2. When should multiselect() be used?

Ans: The multiselect() component should be used when users need to select one or more items from a long, predefined list while minimizing screen space, such as for filters, tags, or assigning multiple categories.

3. What type of data does multiselect() return?

Ans: multiselect() (specifically st.multiselect in Streamlit) returns a list of strings containing the options selected by the user.

4. Can selectbox() have a default value?

Ans: Yes, st.selectbox() in Streamlit can have a default value. We can set it by using the index parameter, which accepts the integer index (starting from 0) of the option we want to be selected by default.

5. How do selection widgets improve user experience?

Ans: Selection widgets improve user experience by streamlining data input, reducing cognitive load, and enabling efficient interaction with digital interfaces.

Date: 4th April 2026

Topic: Streamlit Input Widgets – text_input, number_input

Technology: SQL

1. What is the difference between text_input() and number_input()?

Ans: The core difference between st.text_input() and st.number_input() is the expected input type and widget functionality. text_input() accepts any string, allowing text and symbols, while number_input() restricts input to numerical values (integers or floats), often providing plus/minus arrows for adjustment.

2. How can you limit the values in number_input()?

Ans: In Streamlit, we can limit the values in a `st.number_input()` widget using the `min_value`, `max_value`, and `step` parameters within the function call.

3. What data type does `text_input()` return?

Ans: The `st.text_input()` function in Streamlit returns a string (`str`) data type. When used in SQL queries, this input is treated as character data (such as `VARCHAR`, `TEXT`, or `NVARCHAR`) and must be properly handled to prevent SQL injection, usually by wrapping it in quotes within the SQL string or using parameterized queries.

4. What happens if `text_input()` is empty?

Ans: When a `text_input()` field is empty in SQL-related contexts (like applications or forms), it typically passes an empty string (`""`) or a `NULL` value to the database, depending on the framework, rather than omitting the parameter entirely. This usually results in storing an empty string, allowing a search for blank entries, or potentially causing errors if the column is defined as `NOT NULL`.

5. How do input widgets help in building interactive apps?

Ans: Input widgets (sliders, text inputs, buttons, checkboxes) build interactive apps by enabling real-time user input, allowing users to manipulate data, trigger actions, and customize views instantly without refreshing the page.

Regards
Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>
Date: Sat, 4 Apr 2026 at 17:59
Subject: Fwd: TA || Ashka || TM- B141
To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of Aggregate Functions (SUM, COUNT, AVG), Comparison Operators in SQL, Constraints in SQL, Creating a Database, SQL, Creating a Table, Data Types in SQL, Deleting Data, Foreign Key, GROUP BY, HAVING Clause, Index Basics, INNER JOIN, Inserting Data, Introduction to SQL, Introduction to Views, Joins Overview, LEFT JOIN, Limiting Results, Logical Operators (AND, OR, NOT), NULL Values, Primary Key, Relational Database Concept, Selecting Data, Sorting Data using ORDER BY, Subqueries Basics, Tables, Rows, and Columns, Tables, Rows, and Columns, Types of Databases, Updating Data, What is Database and WHERE Clause. Kindly find the requested details below.

Date: 28th March 2026
Topic: Aggregate Functions (SUM, COUNT, AVG)
Technology: SQL

1. What are aggregate functions?

Ans: Aggregate functions are built-in operations that perform a calculation on a set of values across multiple rows and return a single summarized value.

2. Difference between `COUNT(*)` and `COUNT(column)`?

Ans:

Feature	COUNT(*)	COUNT(column)
What is counted?	All rows in the table/group.	Only rows with non-NULL values in that column.
NULL Handling	Includes NULLs.	Excludes NULLs.
Duplicates	Includes duplicates.	Includes duplicates (unless DISTINCT is used).
Performance	Highly optimized; often uses the smallest available index.	May require a scan of the specific column to check for NULLs.

3. Can SUM work on text?

Ans: No, the standard SQL SUM() function does not work directly on text (e.g., VARCHAR, CHAR, or TEXT data types). It is strictly an aggregate function designed for numeric data types such as INT, DECIMAL, FLOAT, and REAL.

4. What happens if column has NULL values?

Ans: In SQL, a NULL value represents missing, unknown, or inapplicable data. It is a special marker, not a value like zero (0) or an empty string ("").

5. How are aggregate functions used with GROUP BY?

Ans: Aggregate functions are used with the GROUP BY clause to summarize data into meaningful categories.

Topic: Comparison Operators in SQL
Technology: SQL

1. What are comparison operators in SQL?

Ans: SQL Comparison Operators are used to compare two values and filter the result set based on a specific condition.

- Used in the WHERE clause to filter records based on conditions
- Help check for equality, range, or inequality
- Make SQL queries powerful and precise
- Examples: =, >, <, >=, <=

4 attachments

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Projects Link List_Github - Sheet1.pdf
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Projects Link List_Github.xlsx
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